

Mobile Forensic

Module Objectives

- Mobile Phone
- Different Mobile Devices
- Hardware and Software Characteristics of Mobile Devices
- Cellular Network
- Mobile Operating Systems
- Types of Mobile Operating Systems
- What a Criminal Can Do with Mobiles Phones



- Mobile Forensics
- Mobile Forensics Challenges
- Memory Considerations in Mobiles
- Precautions to Be Taken Before Investigation
- Mobile Forensics Process
- Mobile Forensics Software Tools
- Mobile Forensics Hardware Tools



Module Flow



**Mobile
Phones**



**Mobile Operating
Systems**



**Mobile
Forensics**



**Mobile Forensics
Process**



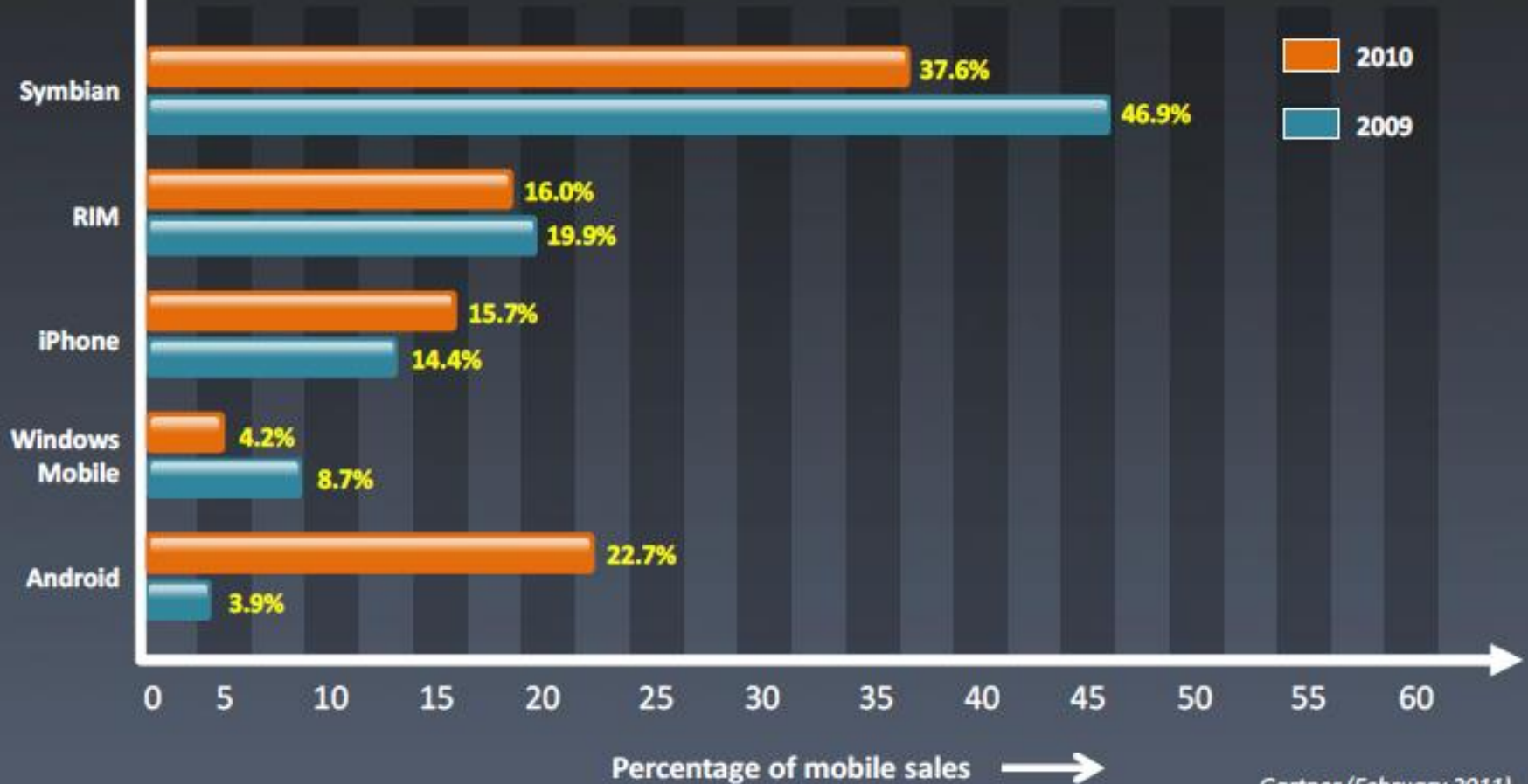
**Mobile Forensics
Software Tools**

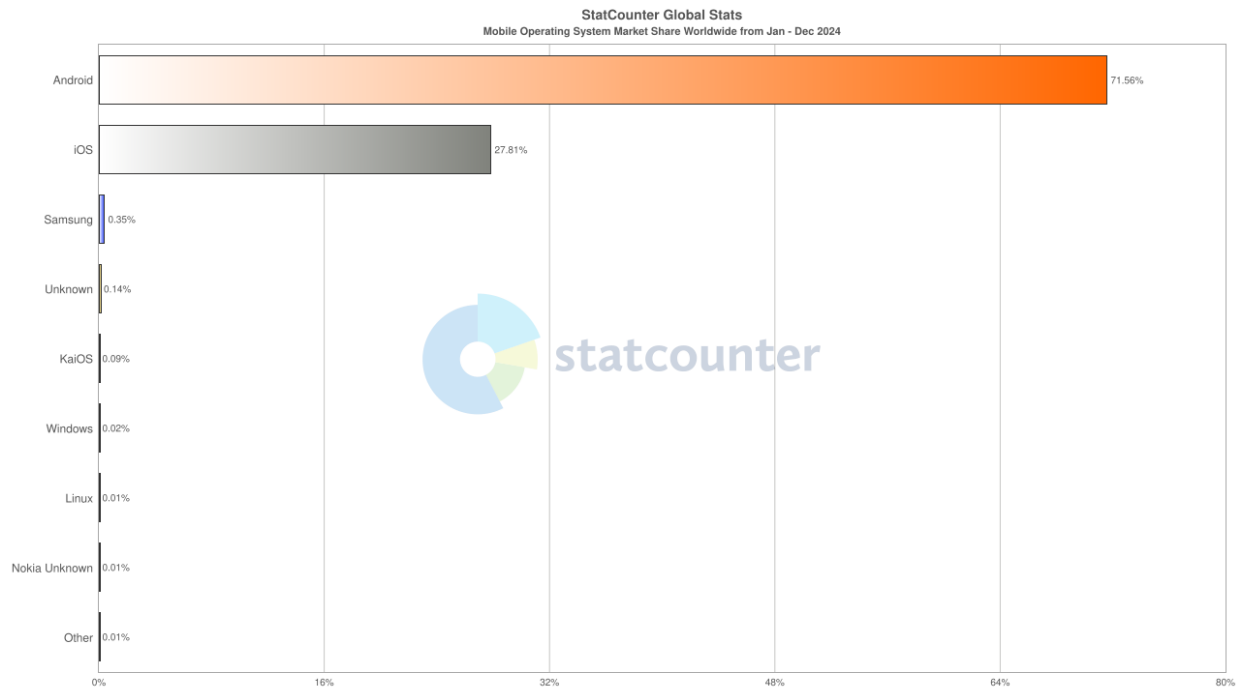
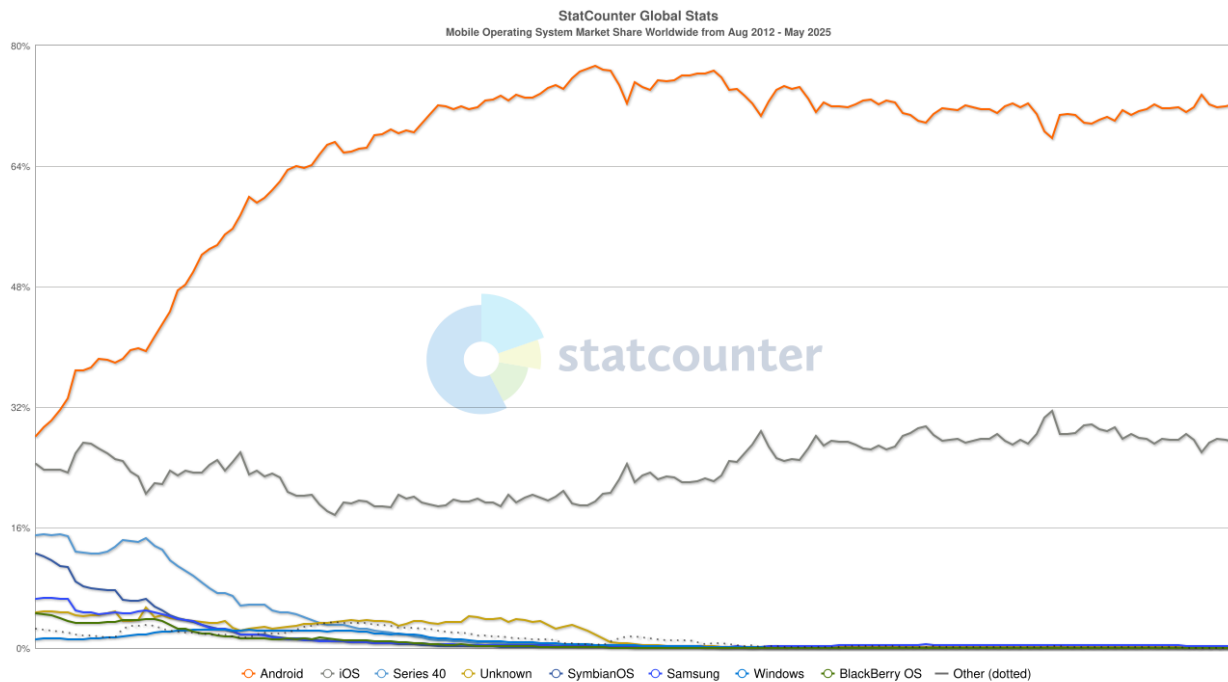


**Mobile Forensics
Hardware Tools**

Smartphone Sales Statistics 2010/2011

172M devices sold in 2009 and 297M devices sold in 2010
Smartphone market increased by 72% during just 1 year





Mobile Phone

The mobile phone or cellular phone is a **complex electronic device** that contains many features



Different Mobile Devices



BlackBerry

- BlackBerry is a **personal wireless handheld device** that supports email, mobile phone capabilities, text messaging, web browsing and other wireless information services
- A BlackBerry can be used as a **phone, address book, or calendar, and to create to-do lists and access wireless Internet**

iPod

- iPod is a **portable digital audio and video player** offering a huge storage capacity

iPhone

- The iPhone is an **Internet-connected multimedia Smartphone** designed and marketed by Apple Inc. with a **multi-touch screen and a minimal hardware interface**

Hardware Characteristics of Mobile Devices

Characteristics	Basic Phone	Advanced Phone	Smartphone
Processor	Limited Speed	Improved speed	Superior Speed
Memory	Limited Capacity	Improved Capacity	Superior Capacity, Built-in Hard Drive Possibility
Display	Grayscale	Color	Large size, 16-bit Color (65,536 colors) or Higher
Card Slots	None	MiniSD or MMCmobile	MiniSD or MMCmobile
Camera	None	Still	Still, Video
Text Input	Numeric Keypad	Numeric keypad, soft keyboard	Touch Screen Handwriting Recognition, Built-in QWERTY style keyboard
Cell Interface	Voice and Limited Data	Voice and High Speed Data	Voice and Very High Speed Data
Wireless	IrDA	IrDA, Bluetooth	IrDA, Bluetooth, WiFi
Battery	Fixed, Rechargeable Lithium Ion Polymer	Removable, Rechargeable Lithium Ion Polymer	Removable, Rechargeable Lithium Ion

Software Characteristics of Mobile Devices

Characteristics	Basic Phone	Advanced Phone	Smartphone
OS	Proprietary	Proprietary	Linux, Windows Mobile, RIM OS, Palm OS , Symbian
PIM	Simple Phonebook	Phonebook and Calendar	Reminder List, Enhanced Phonebook and Calendar
Applications	None	MP3 Player	MP3 Player, Office Document Viewing
Messaging	Text Messaging	Text with Simple Embedded Images and Sounds	Text, Enhanced Text, Full Multimedia Messaging
Chat	None	SMS chat	Instant Messaging
Email	None	Via Network Operator's Service Gateway	Via POP or IMAP Server
Web	None	Via WAP Gallery	Direct HTTP
Wireless	IrDA	IrDA, Bluetooth	IrDA, Bluetooth, WiFi

Components of Cellular Network

1

Mobile Switching Center (MSC): It is the **switching system** for the cellular network



2

Base Transceiver Station (BTS): It is radio transceiver equipment that **communicates with mobile phones**



3

Base Station Controller (BSC): It manages the transceiver's equipment and performs **channel assignment**



4

BSS: Base Station Subsystem is responsible for managing the radio network and is controlled by Mobile service switching center (MSC). It consists of the elements BSC (Base Station controller), BTS (Base Transceiver Station), and TC (Transcoder)

5

Home Location Register (HLR): It is the database at MSC. It is the **central repository system** for subscriber data and service information

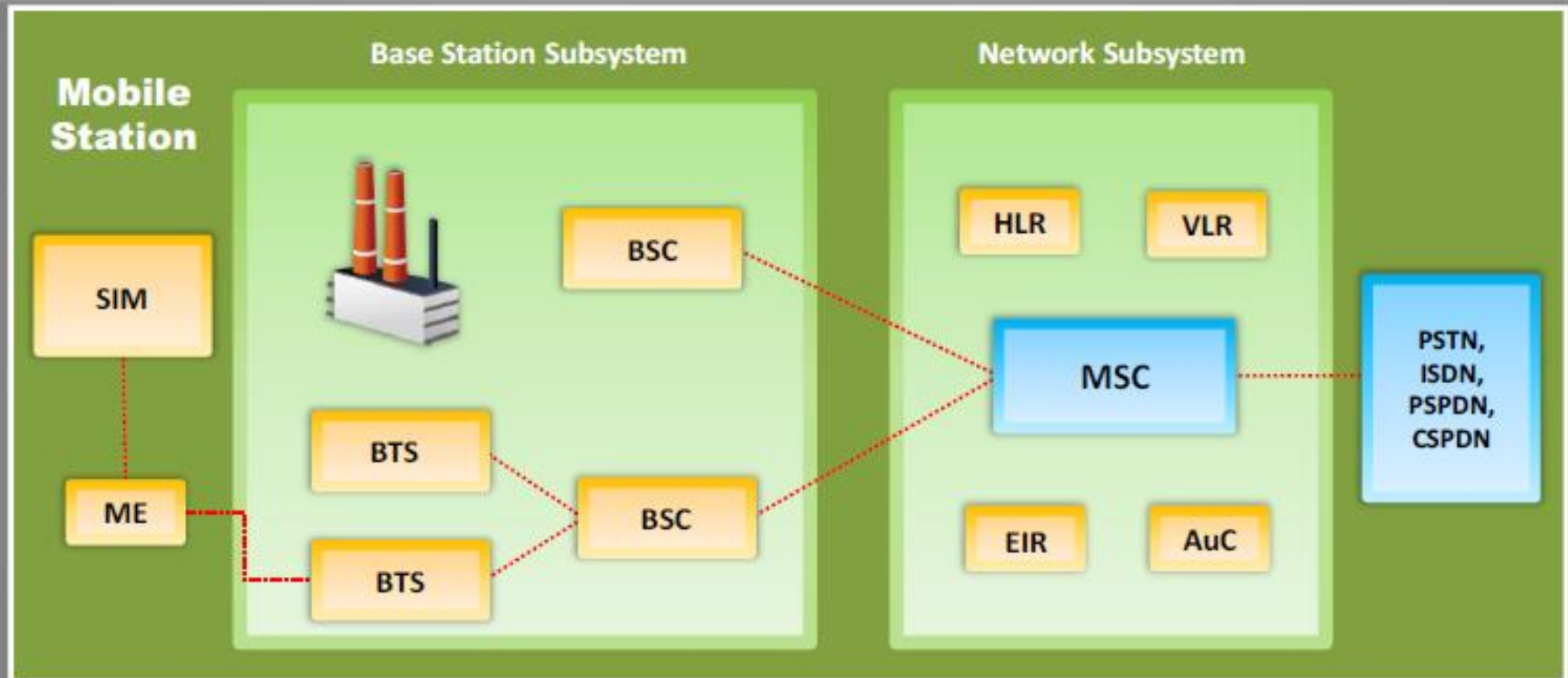


6

Visitor Location Register (VLR): It is the **database** used in conjunction with the HLR for mobile phones roaming outside their service area



Cellular Network



SIM: Subscriber Identity Module

MSC: Mobile Services Switching Center

HLR: Home Location Register

BTS: Base Transceiver Station

AuC: Authentication Center

VLR: Visitor Location Register

BSC: Base Station Controller

ME: Mobile Equipment

EIR: Equipment Identity Register

Module Flow



**Mobile
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**Mobile Operating
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**Mobile
Forensics**



**Mobile Forensics
Process**



**Mobile Forensics
Software Tools**



**Mobile Forensics
Hardware Tools**

Mobile Operating Systems

1

A mobile operating system is the **operating system** that operates a mobile device like a mobile phone, smartphone, PDA, etc.



2

It determines the **functions and features** available on mobile devices such as keyboards, applications, email, text messaging, etc.



3

It **manages the communication** between the mobile device and other compatible devices like computers, televisions, or printers



It controls which **third-party applications** can run on the device



4

Types of Mobile Operating Systems

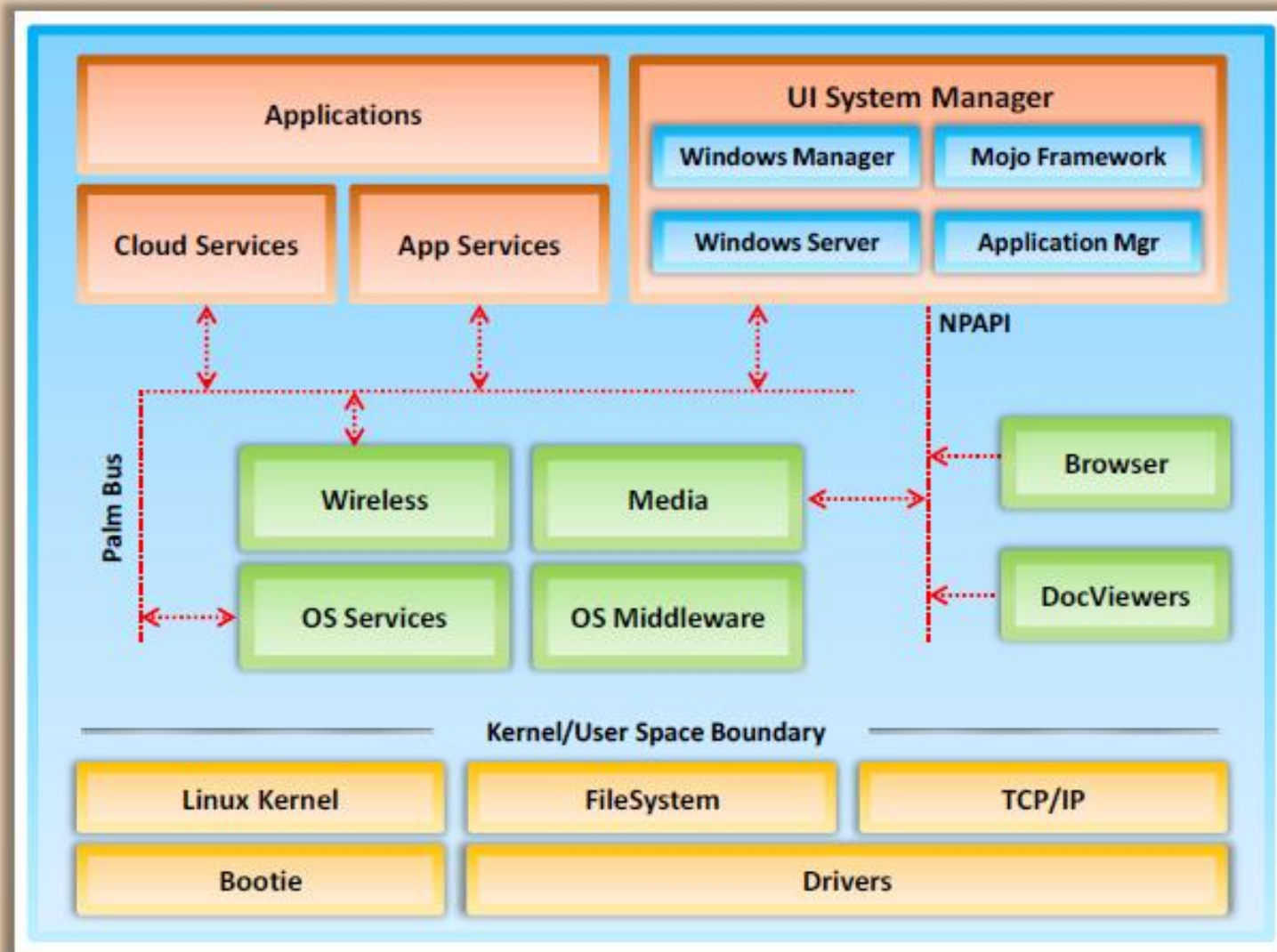


webOS

- webOS is a **mobile operating system** running on the **Linux kernel**
- It uses **multi-touch gestures** to navigate on the **touchscreen**
- It allows users to access Gmail, Yahoo!, Facebook, LinkedIn, etc., using its **Synergy** feature, and integrates all these sources into a single list
- It uses the **WebKit layout engine** to access web browsers



webOS System Architecture



Symbian OS

- Symbian OS is a **mobile operating system** designed for smartphones and currently maintained by Nokia
- It is designed to make minimal demands on batteries and to have **low memory**



Associated libraries



User interface frameworks



Memory management optimized for
embedded software environment



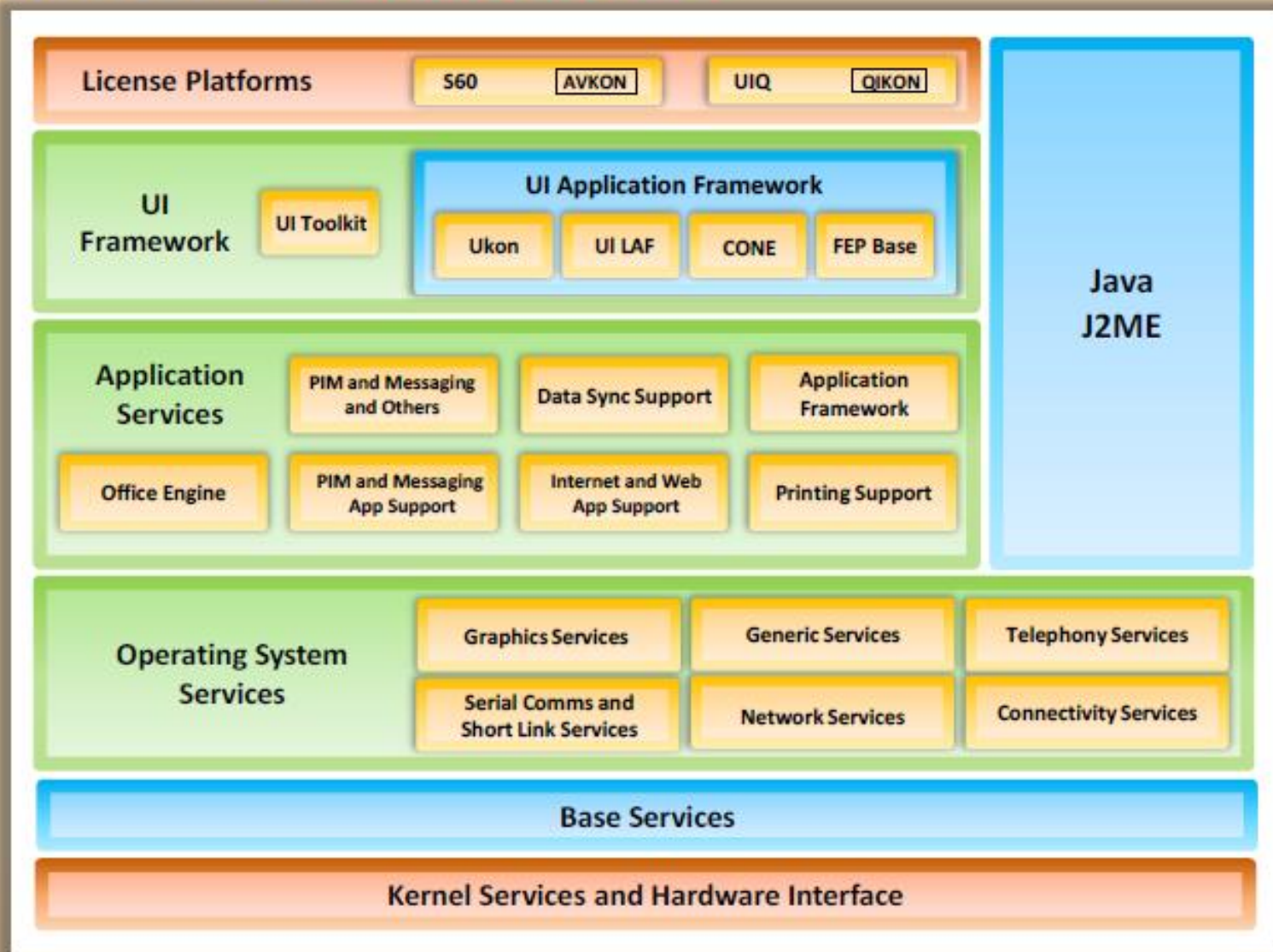
WebKit-based browser



Object-oriented software architecture

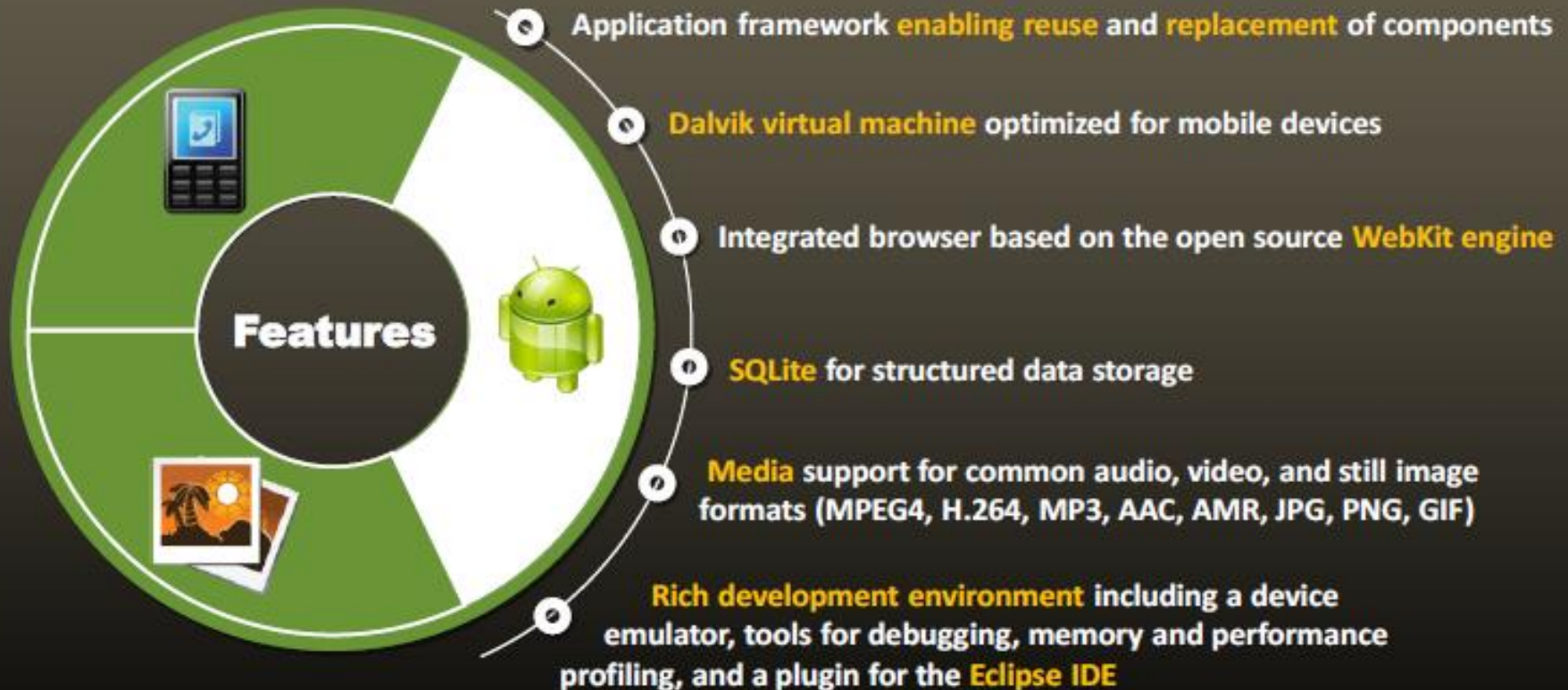


Symbian OS Architecture



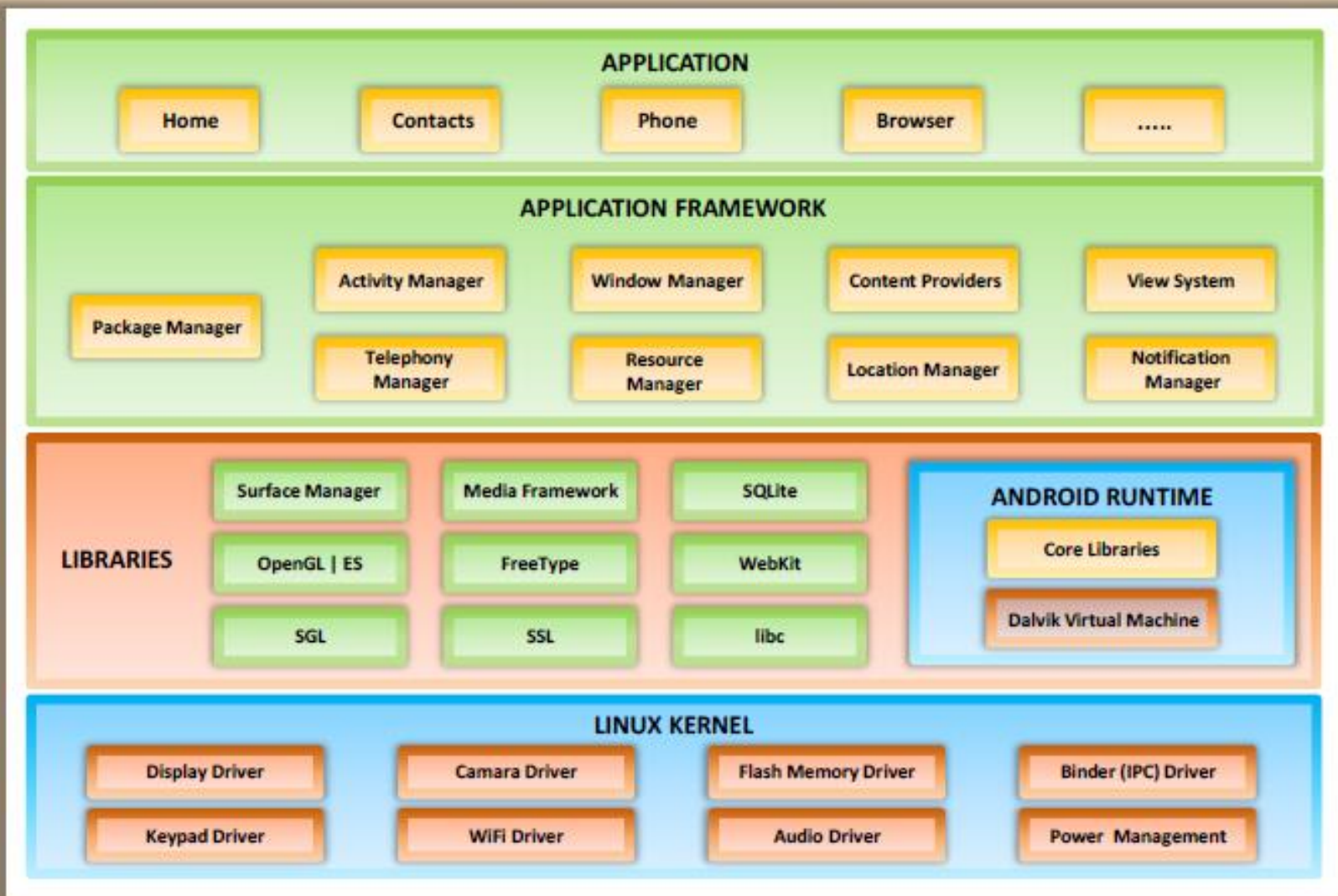
Android OS

Android is a software stack developed by **Google for mobile devices** that includes an operating system, middleware, and key applications



<http://developer.android.com>

Android OS Architecture



RIM BlackBerry OS

- BlackBerry OS is a mobile operating system developed by **Research In Motion (RIM)** for its BlackBerry line of **smartphone** handheld devices
- It provides **multitasking** and supports particular input devices used in BlackBerry such as trackwheel, trackball, trackpad, and touchscreen





Windows Phone 7

- Windows Phone 7 is a mobile operating system developed by **Microsoft**, and is the successor to its **Windows Mobile platform**
- Windows Phone 7 features:



Smooth transitional user interface



Good Office integration and support



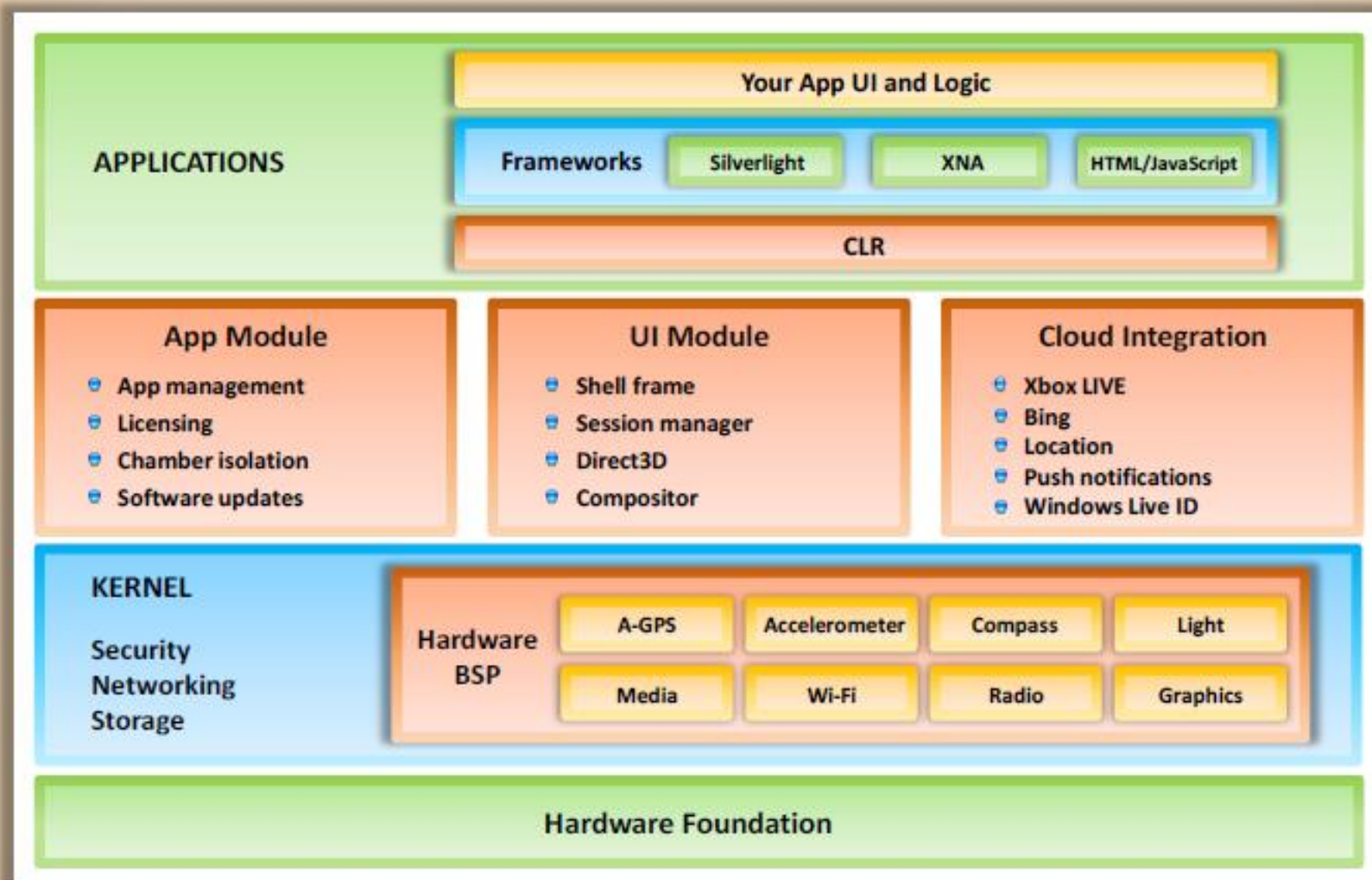
Easy setup for Google, Windows Live, Yahoo! Mail, and others



Informative lock screen and auto-hidden top status bar

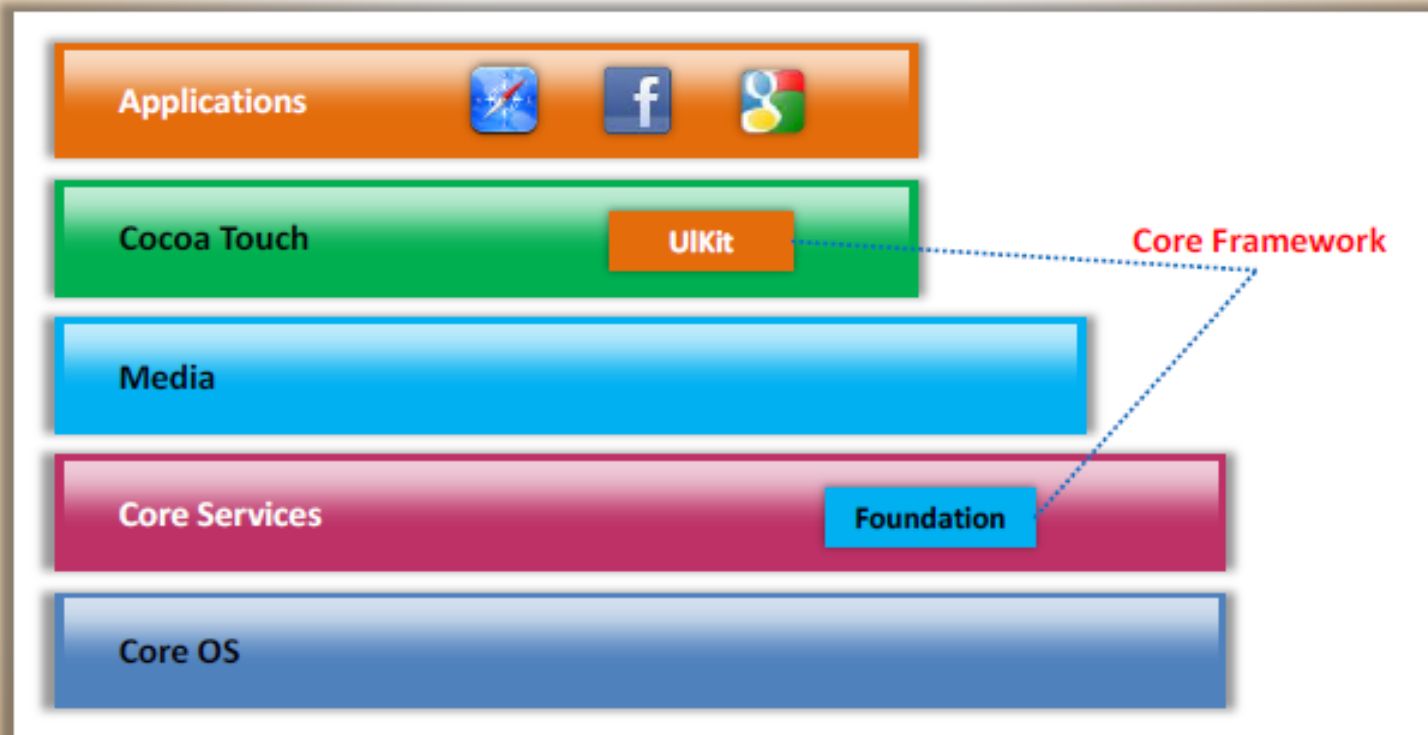


Windows Phone 7 Architecture



Apple iOS

- iOS is **Apple's mobile operating system**, which supports Apple devices such as iPhone, iPod touch, iPad, and Apple TV
- The user interface is based on the concept of **direct manipulation**, using **multi-touch** gestures



Module **Flow**



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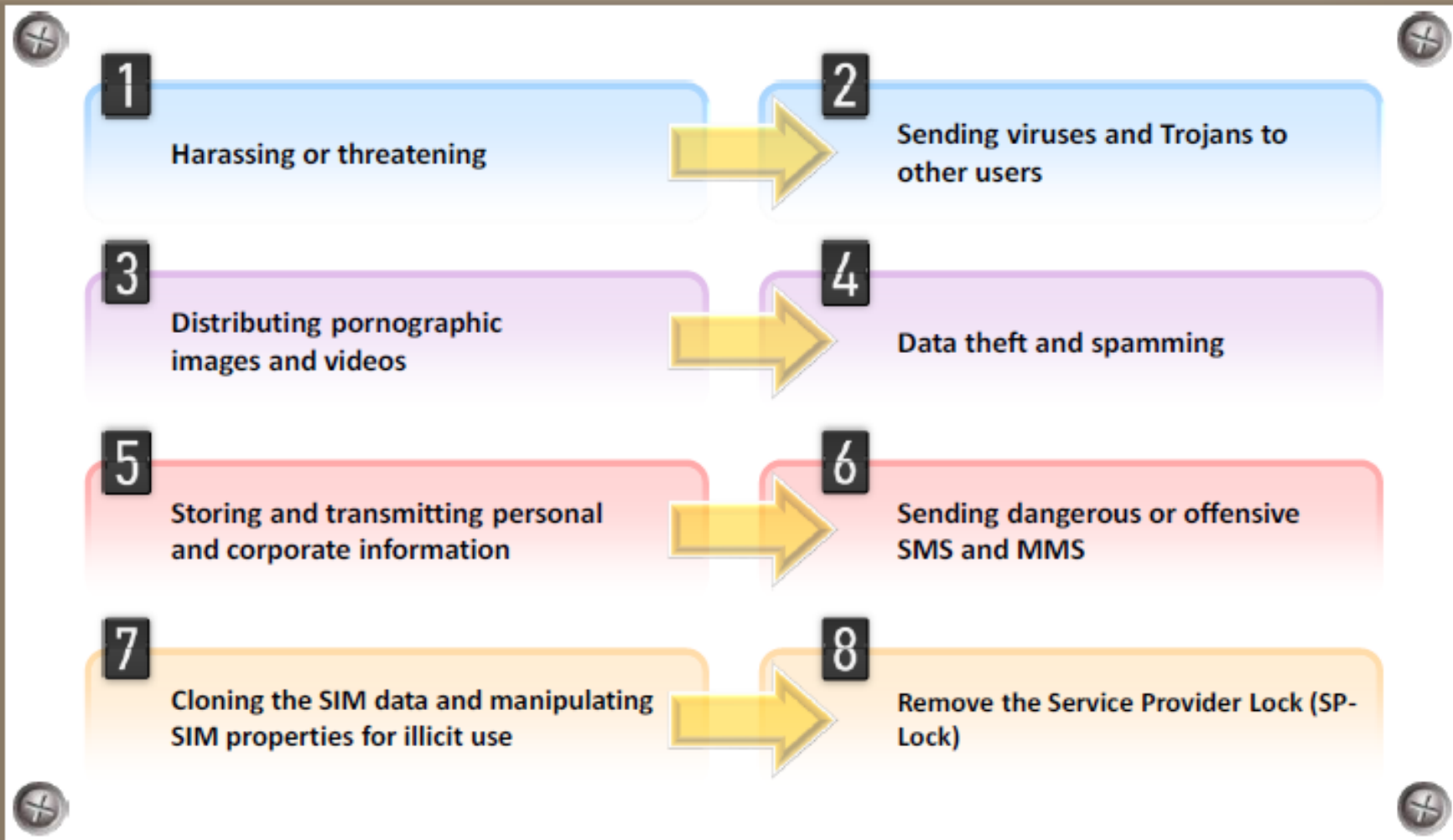


**Mobile Forensics
Software Tools**



**Mobile Forensics
Hardware Tools**

What a Criminal Can Do with **Mobiles Phones**



Mobile Forensics



Mobile phone forensics is the **science of recovering digital evidence** from a mobile phone under forensically sound conditions



It includes recovery and analysis of data from **mobile devices** and **SIM cards**



Mobile forensics aim to catch the **perpetrators** of crimes that involve the use of **mobile phones**

Mobile Forensics Challenges

Often a disposable solution for criminals



Devices are not widely supported by forensic solutions



No contract and no identity tied to the device or service contract



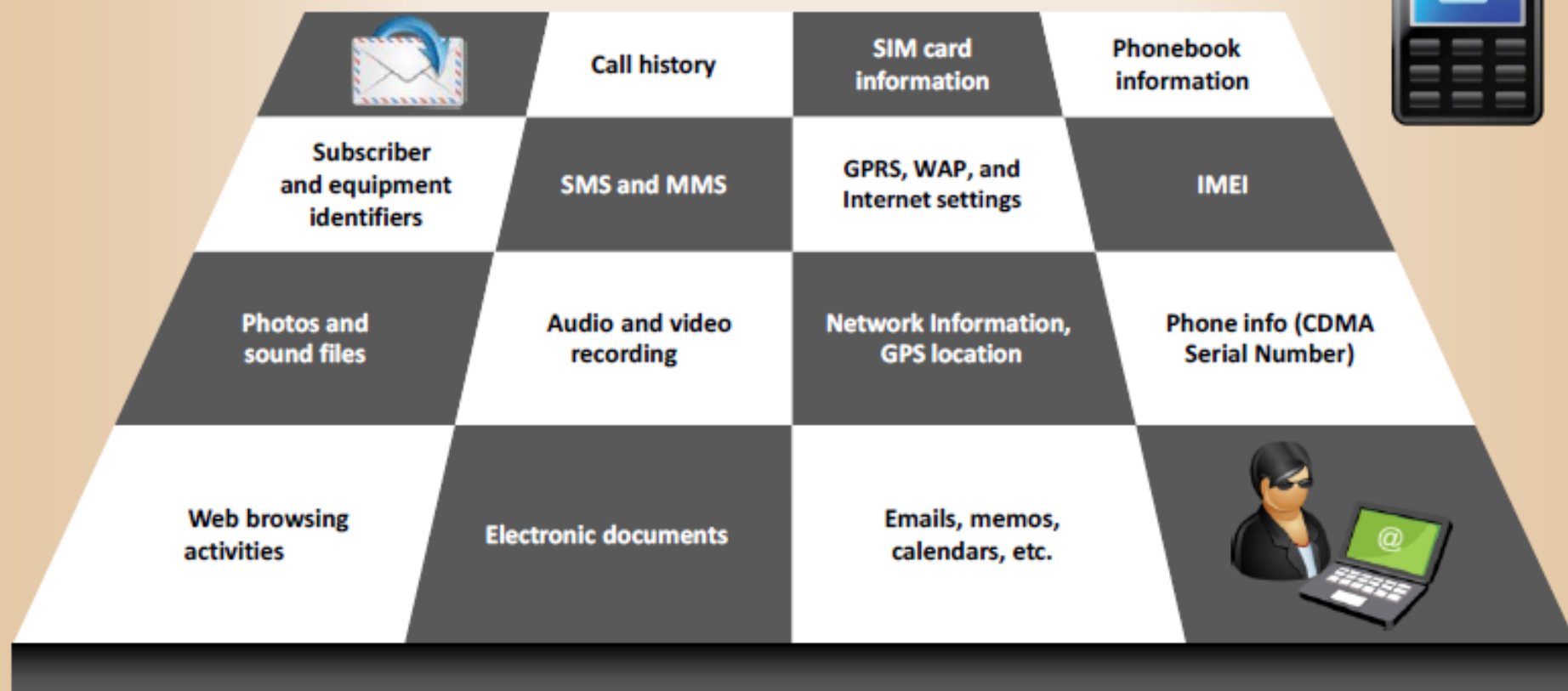
No single standardized approach to investigate mobile devices

Various forensic tools are only able to operate on a particular handset, specific platforms for a specific product, a distinct operating system, or specific hardware architecture



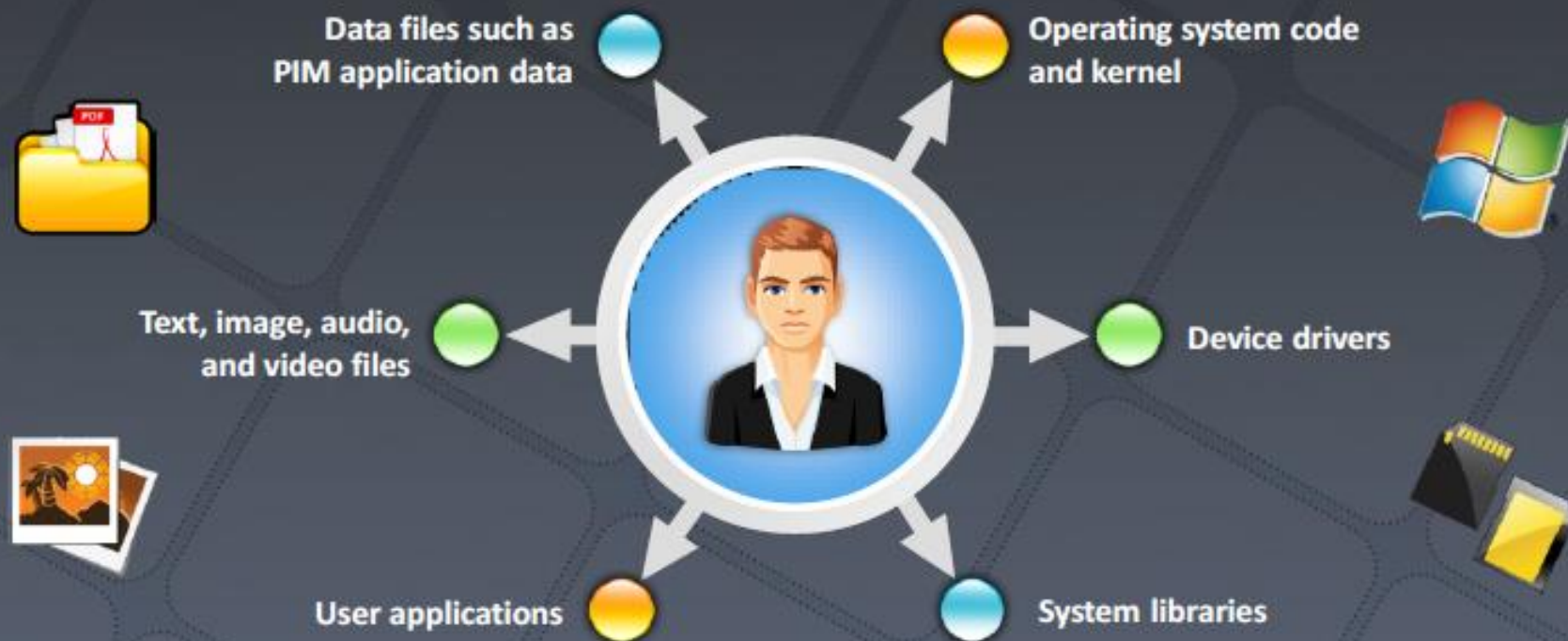
Ever-changing advancement of mobile devices increases the complexity of mobile device examinations

Forensics Information in Mobile Phones



Memory Considerations in Mobiles

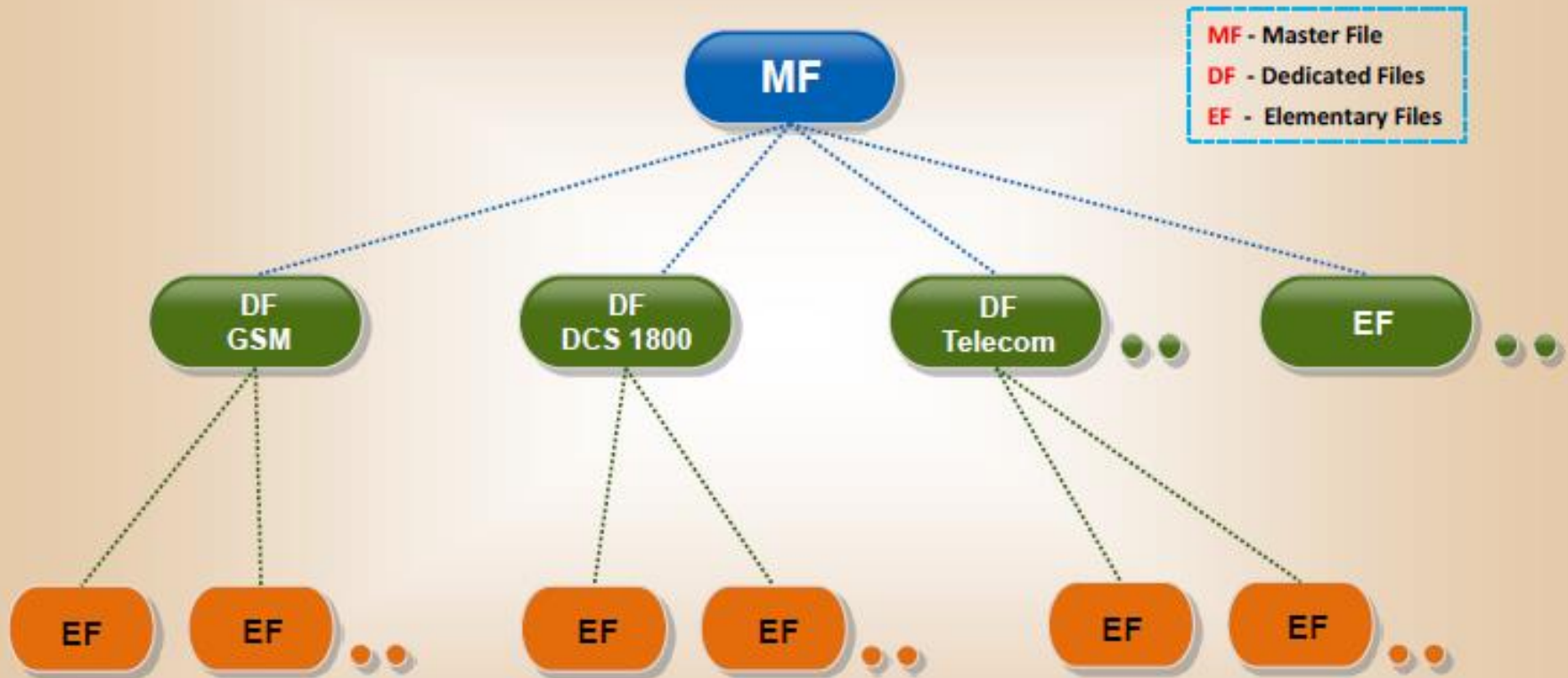
- A mobile phone contains various types of **volatile** and **nonvolatile** memory
- It stores several kinds of data, including:



Subscriber Identity Module (**SIM**)



SIM File System



International Mobile Equipment Identifier (**IMEI**)

- IMEI is a **15-digit number** that indicates the manufacturer, model type, and country of approval for GSM devices
- First eight digits, known as the **Type Allocation Code (TAC)**, give the model and origin
- For powered on GSM and UMTS phones, the International Mobile Equipment Identifier (IMEI) can be obtained by keying in ***#06#**



Precautions to Be Taken Before Investigation



If the phone is in a cradle or connected to a PC with a cable, then **seize the phone with the cable and cradles**, because unplugging the device from the computer may **eliminate the data transfer or overwrite the synchronization**

If the phones are found in a compromised state such as immersed in a liquid, **remove the battery to prevent electrical shorting** and seal the remainder of the mobile phone in a proper container filled with the same liquid, which should not be caustic



Isolate the phone from the radio network, which **helps to keep new traffic from overwriting the existing data**

Isolate the phone from other synchronized devices, which **keeps the new data from affecting the existing data**



Some mobile communication devices use alkaline batteries as a power source; **replace such batteries in transit to minimize the risk of data loss** due to complete battery discharge

Module **Flow**



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Mobile Forensic Process



Collecting the Evidence

Protect the integrity of **traditional and electronic evidence**

Prevent **unauthorized users** from entering at the scene and touching the evidence

Collect all the electronic **devices found at the crime scene**

Check whether the mobile device is **connected to a computer**

Confirm the **power state of the devices** by checking flashing light

Collect **non-electronic evidence** such as written passwords, handwritten notes, and computer printouts



Points to Remember while Collecting the Evidence

If the device is on, do **not** turn it off

- Turning it off could **activate lockout feature**
- Write down all **information** on display (photograph if possible)
- Keep it charged and **protect it from tampering**
- Do not press any key, it **may lose the data** in the device



If the device is off, leave it off

- Turning it on could **alter evidence** on device
- **Do not remove the battery**; this may cause the contents of some devices to be lost



Collecting an iPod/iPhone Connected to a Computer

If an iPod/iPhone is connected to a Mac computer

- Check whether the **device is mounted** by looking at the screen
- If it is mounted, **eject the device by dragging the icon** of the iPod/iPhone to the trashcan on the Macintosh desktop



If an iPod/iPhone is connected to a Windows computer

- Note the name of the iPod/iPhone and check whether **the device is mounted or not**
- If it is mounted, eject the device by clicking the **"Unplug or eject hardware"** icon on the task bar



Document the **Scene** and Preserve the **Evidence** (Cont'd)



1. Document all the **electronic devices** found at the crime scene
2. Take photographs of all evidence at the scene and **write notes on what you have seen** on the screen
3. Document the **state of the device** during seizure
4. Document any **activity on the electronic devices** found at the crime scene
5. Handle electronic evidence in a manner that **preserves its integrity**
6. Protect the electronic evidence from magnetic fields, dust, vibration, and other factor that may **damage the integrity** of the electronic evidence
7. Secure the devices from **mechanical or electrical shock**



Document the Scene and Preserve the Evidence

Preserve all the evidence and documents in a **secure location**

Focus on hidden or trace evidence and take **necessary actions to preserve it**

Pack the electronic devices in **antistatic packaging**

Make sure that all the containers that hold the **evidence are labeled** in an appropriate way

Keep electronic evidence away from **magnetic sources** while transporting

Store the evidence in a secure area and weather controlled environment that is **away from high temperature and humidity**

Maintain the **chain of custody** documents

Imaging and Profiling

- Imaging is the process of **creating an exact copy of the contents** of a digital device to prevent the accidental modification of the original evidence
- Use data imaging tools, which make **exact bit-to-bit copies of the originals** and prevent any alteration



Acquire the **Information**



Acquisition of data at the scene avoids **loss of information** due to battery depletion, damage during transportation and storage

Data acquisition process at the crime scene is hampered due to **lack of controlled settings**, appropriate equipment, and other prerequisites

Try to acquire the data from **images of the evidence** such as SIM cards, memory cards, or directly from the mobile device itself

Use **data acquiring tools** such as SIM Card Data Recovery and SIMCon to recover the data from evidence, SIM cards, or memory cards

Device Identification

- Mobile **device needs to be identified** by the model, operating system, and service provider
- This information allows examiners to **choose the proper data acquisition tool** to collect the evidence



Check the following for **device** identification:

- Check the **battery cavity** to find manufacturer's name
- If the mobile device is powered on, **check the screen to find the type of phone**, service provider name, and operating system used
- If the mobile device is powered off, remove the battery and check the manufacturer logo to **find the model number, IMEI, and FCC ID**

Acquire Data from **SIM Cards** (Cont'd)

SIM contains important information related to the forensics investigation:



Service related information such as **unique identifiers** for the (U)SIM, the **Integrated Circuit Card Identification (ICCID)**, and the subscriber, the **International Mobile Subscriber Identity (IMSI)**

Phonebook and call information such as Abbreviated **Dialing Numbers (ADN)** and **Last Numbers Dialed (LND)**



Messaging information including **SMS**, **EMS**, and **multimedia** messages

Location information, including **Location Area Information (LAI)** for voice communications and **Routing Area Information (RAI)** for data communications



Acquire Data from SIM Cards



To access the SIM, **PIN code** (Personal Identification Number) is required



Failure to enter a valid PIN in three attempts blocks the card and then an **eight-digit PUK** (Personal Unlock Number) must be entered



PUK is provided by the **network operator** and cannot be changed by the user



Failure to get correct PUK in 10 attempts **disables** the SIM permanently



Investigator should ask the network operator for PUK **to gain access** to the SIM



Use SIM card data recovery tool such as **SIM Analyzer, SIMCon**, etc., to recover data

Acquire Data from **Unobstructed** Mobile Devices



An unobstructed device does **not require a password** or other authentication technique to access the device and perform an acquisition



Unobstructed devices include mainly **CDMA phones**, **freestanding (U)SIMs**, and **GSM phones** containing a (U)SIM



Record the **time** and **date** in the phones



Check with the contacts, SMS, and other entries



Use different data recovery tools such as **Cell Phone Analyzer** to recover the deleted information from the device

Acquire the Data from **Obstructed Mobile Devices**



- Obstructed devices typically refer to devices that are **shut off** and require successful authentication to **gain access**

- Recover the information from such devices using the following techniques:

- Ask the victim or suspect for **PIN**
- Review the seized **non-electronics materials** such as notes or print outs
- Contact the **service provider**
- Contact the device **manufacturer** and **service provider** for information on known backdoors and vulnerabilities that might be exploited
- Contact the device maintenance and repair companies, as well as commercial organizations that provide architecture information on handheld device products
- Use different forensics tools such as **Cell Phone Analyzer**
- Use a data recovery tool such as **SIM Analyzer** and **SIMCon**



Acquire Data from **Memory Cards** (Cont'd)



Removable media **extends the storage capacity** of mobile phones, allowing individuals to store additional files beyond the device's built-in capacity and to share data between compatible devices

1



Mobile phone supports **Secure Digital (SD)**, **MultiMedia Cards (MMC)**, and other types of **removable media** containing significant amounts of data

2



Recover the data from removable media and memory cards with the use of a **media reader** and a **Memory Card Data Recovery tool**

3

Acquire Data from **Memory Cards** (Cont'd)

Name	Characteristics
Compact Flash Card (CF)	Matchbook size (length-36.4mm, width-42.8mm, thickness-3.3mm for Type I cards and 5mm for type II cards) 50-pin connector, 16-bit data bus
MMCplus (compatible with original MultiMedia Card or MMC)	Postage stamp size (length-32mm, width-24, and thickness-1.4mm) 13-pin connector, 1, 4, or 8 bit data bus (7-pin connector, 1-bit data bus, MCC compatibility)
MMCmobile (compatible with original Reduced Size MMC or RS-MMC)	Thumbnail size (length-18mm, width-24mm, and thickness-1.4 mm) 13-pin connector, 1, 4, or 8 bit data bus (7-pin connector, 1-bit data bus, RS-MMC compatibility) Requires a mechanical adapter to be used in a full size MMCplus slot
MMCmicro	Contact lens size (length-14 mm, width-12 mm, and thickness-1.1 mm) 10-pin connector and a 1 or 4-bit data bus Requires a mechanical adapter to be used in a full size MMCplus slot
Secure Digital (SD) Card	Postage stamp size (length-32 mm, width-24 mm, and thickness-2.1 mm) 9-pin connector, 1 or 4-bit data bus Features a mechanical erasure-prevention switch

Acquire Data from **Memory Cards**

Name	Characteristics
MiniSD Card	Thumbnail size (length-21.5 mm, width-20mm, and thickness-1.4 mm) 9-pin connector, 1 or 4 bit data bus Requires a mechanical adapter to be used in full size SD slot
MicroSD (formerly Transflash)	Contact lens size (length-15 mm, width-11 mm, and thickness-1 mm) 6-pin connector, 1 or 4-bit data bus Requires a mechanical adapter to be used in a full size SD slot
Memory stick	Chewing gum stick size (length-50 mm, width-21.45 mm, thickness-2.8 mm) 10-pin connector, 1-bit data bus Features a mechanical erasure prevention switch
Memory Stick Duo	Partial chewing gum stick size (length-31mm, width-20 mm, thickness-1.6 mm) 10-pin connector, 4-bit data bus Features a mechanical adapter to be used in a full size Memory Stick slot
Memory Stick Duo	Contact lens size (length-12.5 mm, width-15 mm, and thickness-1.2 mm) 11-pin connector, 4-bit data bus Requires a mechanical adapter to be used in a full size Memory Stick slot

Acquire Data from **Synched** Devices

Mobile phones are generally synched with the computer to **save the data as another backup copy**



A significant amount of evidence on a mobile phone may also be present on the **suspect's laptop or personal computer**



Search for types of evidence including **contacts, SMS, email details, images, and videos**



Gather Data from **Network Operator**

- Gather the detailed information from the network operator including **calls made/received**, **message traffic**, **data transferred**, and **connection location/timing**

Home Location Register (HLR) provides:

- Customer's name and address
- Billing name and address (if other than customer)
- User's name and address (if other than customer)
- Billing account details
- Telephone number (MSISDN)
- IMSI
- SIM serial number (as printed on the SIM-card)
- PIN/PUK for the SIM
- Subscriber services allowed



Check **Call Data Records** (CDRs)

CDR files created in the MSC and records information about:

Originating MSISDN

Initial serving Base Station (BTS)

Terminating MSISDN

Connection time

Originating and terminating IMEI

Time the call was disconnected and reason

Time[26]	Qualifier[11]	Calling Number[21]	Called Number[21]	PRI ID Number[5]	B-Channel[3]	Time[26]	Text[35]
Orig time	Call Rqst	calling	called	PRI id	B-channel		
Disc time	Call Disc	calling	called	PRI id	B-channel	connect time	Cause
Orig time	Setup Fail	calling	called	PRI id	B-channel		
Disc time	Disc Fail	calling	called	PRI id	B-channel	connect time	Cause

Gather Data from SQLite Record (Cont'd)

- Mobile operating systems iOS and Android use SQLite databases to store vital information such as **contacts**, **SMS**, and **call records**
- Use **Base Mac SQLite editor** to see the **schema of the message table**



Schema Data SQL Log			
Name	Type	Constraints	
ROWID	INTEGER		
address	TEXT		
date	INTEGER		
text	TEXT		
flags	INTEGER		
replace	INTEGER		
svc_center	TEXT		
group_id	INTEGER		
association_id	INTEGER		
height	INTEGER		
UIFlags	INTEGER		
version	INTEGER		
subject	TEXT		
country	TEXT		
headers	BLOB		
recipients	BLOB		
read	INTEGER		

Gather Data from SQLite Record

- Look for the records in Hex viewer to **get the information** about record length, key, record header length, address length, telephone number, date and time stamp, message length, flag, and country



Diagram illustrating the structure of an SQLite record in a hex viewer, with labels pointing to specific fields:

- Key**: Points to the first byte (0).
- Header Length**: Points to the next two bytes (6E2F).
- Address Length**: Points to the next two bytes (1300).
- Date/Time Stamp**: Points to the next four bytes (27040100).
- Message Length**: Points to the next four bytes (01010001).
- Flag**: Points to the next byte (01).
- Record Length**: Points to the first byte (0).
- Country**: Points to the first byte (0).
- Phone Number**: Points to the last four bytes of the message (+9715020100).

The hex viewer shows the following data for the file **SUSPECT_SMS.bin**:

Offset	Hex	ASCII
0	00	n/...'.A.....
16	00110000	+9715020100
32	3038404E	00MNYUSheikh zay
48	65642072	ed road.Emmar sq
64	75617265	uare.building #
80	340A3674	4.6th floor.TRA
96	6F666669	office.....jo.
112		

Analyze the **Information** (Cont'd)

- Subscriber and equipment identifiers
- Date/time, language, and other settings
- Phonebook information
- Appointment calendar information
- Text messages
- Dialed, incoming, and missed call logs

- Electronic mail
- Photos
- Audio and video recordings
- Multi-media messages
- Instant messaging and web browsing activities
- Electronic documents
- Location information



Analyze the **Information**



1. Identify the individuals who created, modified, or accessed a file
.....
2. Determine when events occurred by analyzing call logs, the date/time, and content of messages and email
.....
3. Create the timeline of the events
.....
4. Recover the hidden information
.....
5. If the entries such as SMS, contacts, emails, etc., are encrypted, then use cryptanalysis tools such as Crank
.....
6. Use password cracking tools such as Hydra to read the password protected information
.....
7. Try to find out the geographical location of the attacker

Generate Report

- Report generation is the **process of preparing a comprehensive summary** of all the steps performed and conclusions reached in the forensics investigation
- A good report relies on proper **documentation, notes, photographs, sketches, and software-generated content**
- Report generated by the forensics tool include **items from the case file**, such as case number, date, title, evidence category, outcome, etc.
- Investigator needs to include **only related findings** in the final report to minimize its size and lessen confusion for the reader



- Mobile forensics report contains the following information:

- Reporting agency name
- Case number
- Case investigator name
- Case submitter name
- Date of receipt and report
- List of items submitted for examination
- Examiner name and signature
- Equipment and setup used in the examination
- Brief description of steps taken during examination
- Details of findings
- Report conclusions

Module Flow



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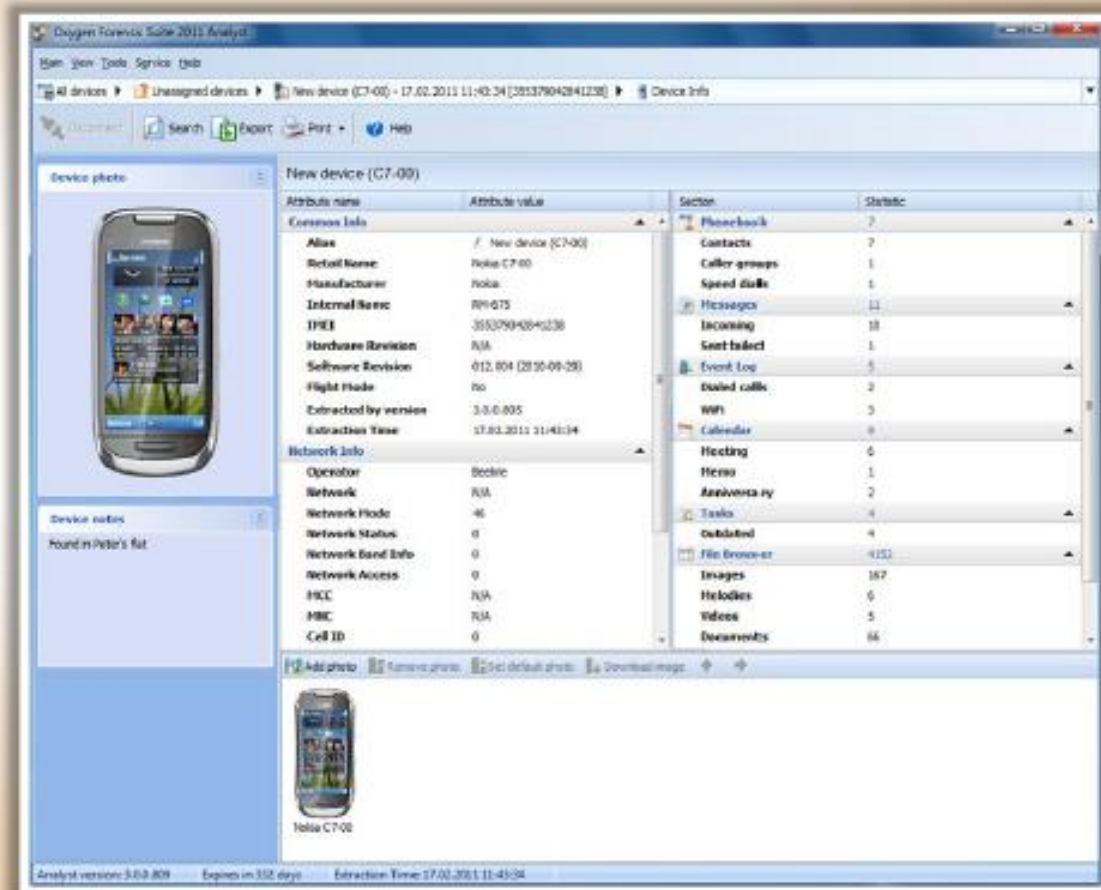
**Mobile Forensics
Hardware Tools**

Oxygen Forensic Suite 2011

Oxygen Forensic Suite 2011 is mobile forensics software that **recovers data from cell phones, smartphones, and PDAs**

Features

- Supports Symbian OS, Apple iPhone, Android, Windows Mobile, and RIM BlackBerry devices
- Stores messages in custom folders and subfolders
- Extracts all files from phone memory as well as from flash card, including installed applications and their data



<http://www.oxygen-forensic.com>

MOBILedit! Forensic

MOBILedit! Forensic **collects all possible data from the mobile phone** and generates an extensive forensics report

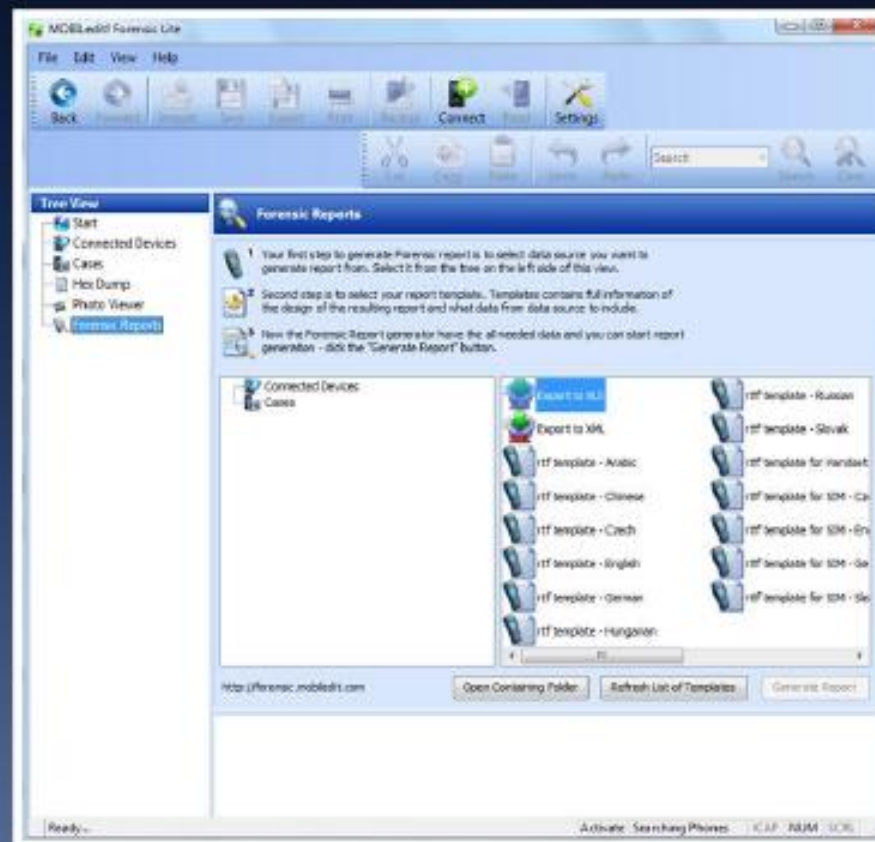
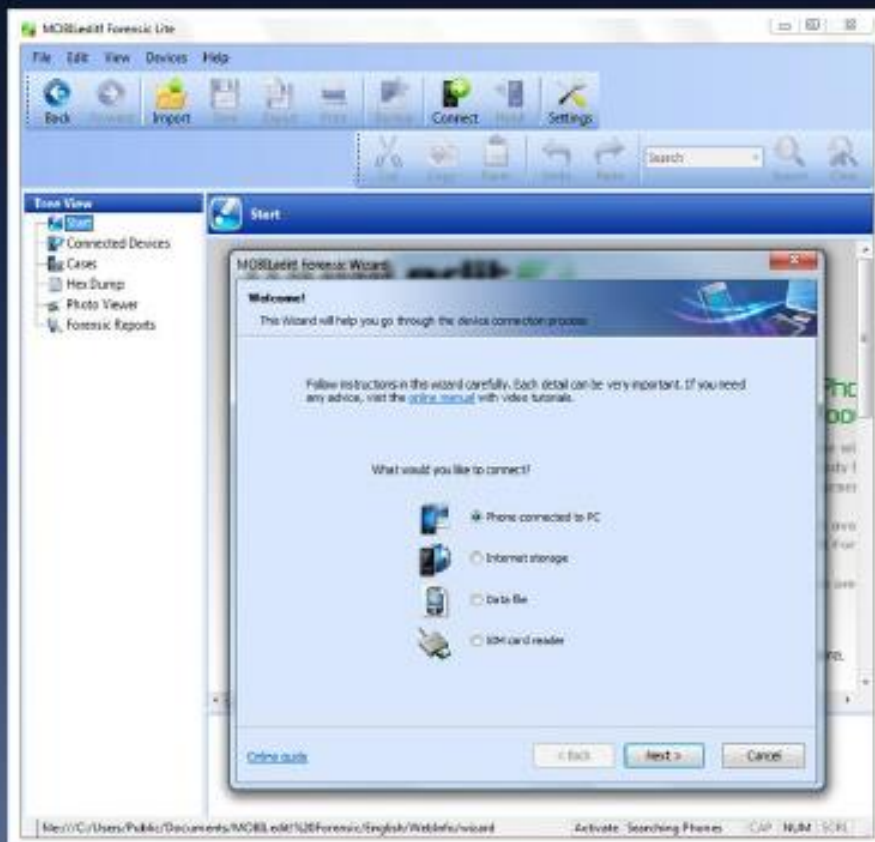


Features

- 1 Retrieves **all data** from a phone with one click
- 2 Reads **deleted messages** from the SIM card
- 3 Generates **forensic reports** ready for the courtroom in a variety of formats
- 4 Allows investigator to manually look through the **contents of the phone and back up data** to the PC
- 5 Direct **SIM analyzer** through SIM readers



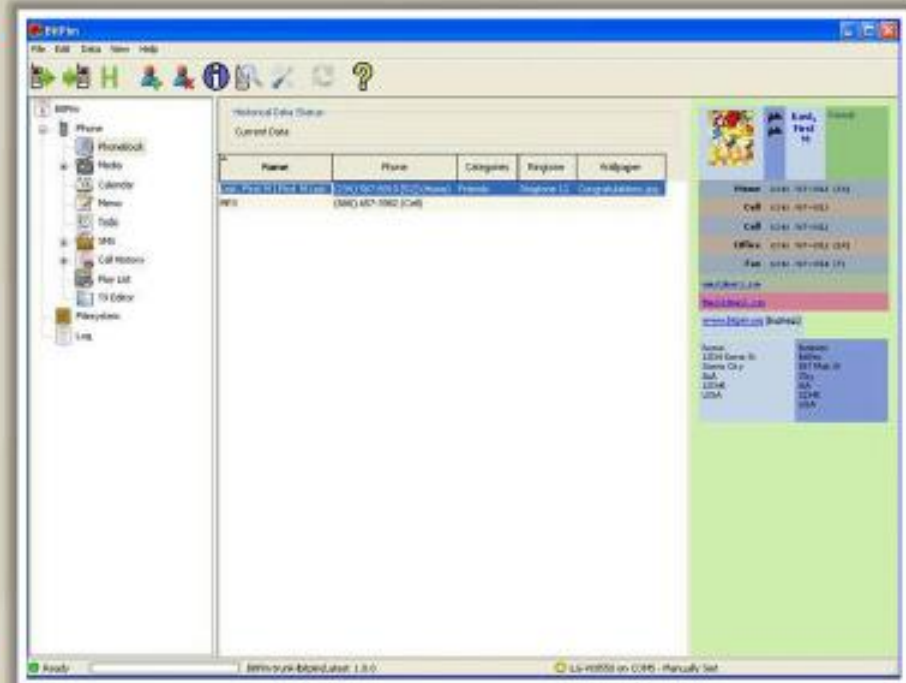
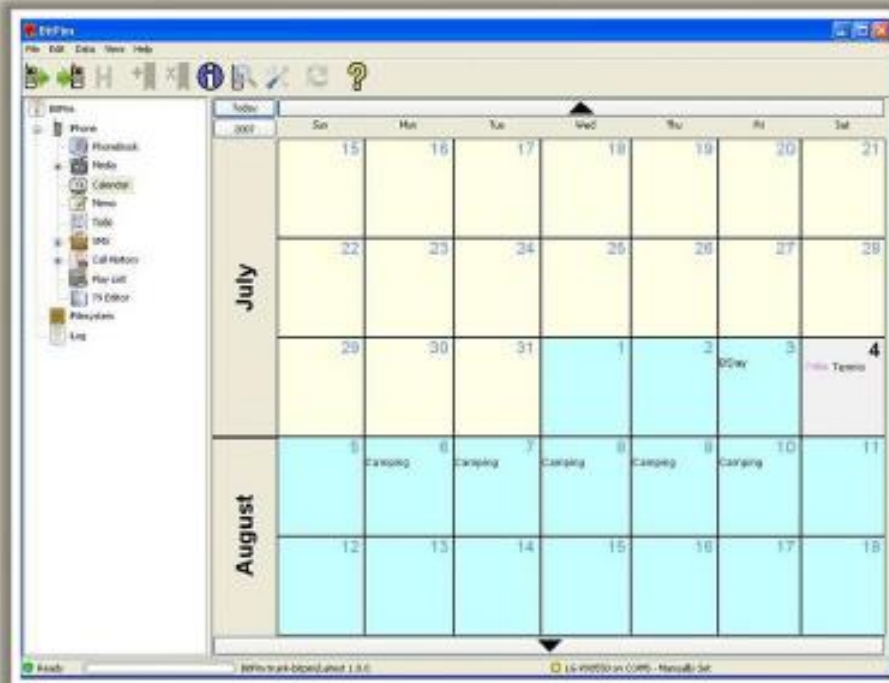
MOBILedit! Forensic: Screenshot



<http://www.mobiledit.com>

BitPim

- BitPim is a program that allows you to **view and manipulate data** on many CDMA phones from LG, Samsung, Sanyo etc.
- This data includes the Phonebook, Calendar, Wallpapers, Ringtones, and the File system for most **Qualcomm CDMA chipset-based phones**



<http://www.bitpim.org>

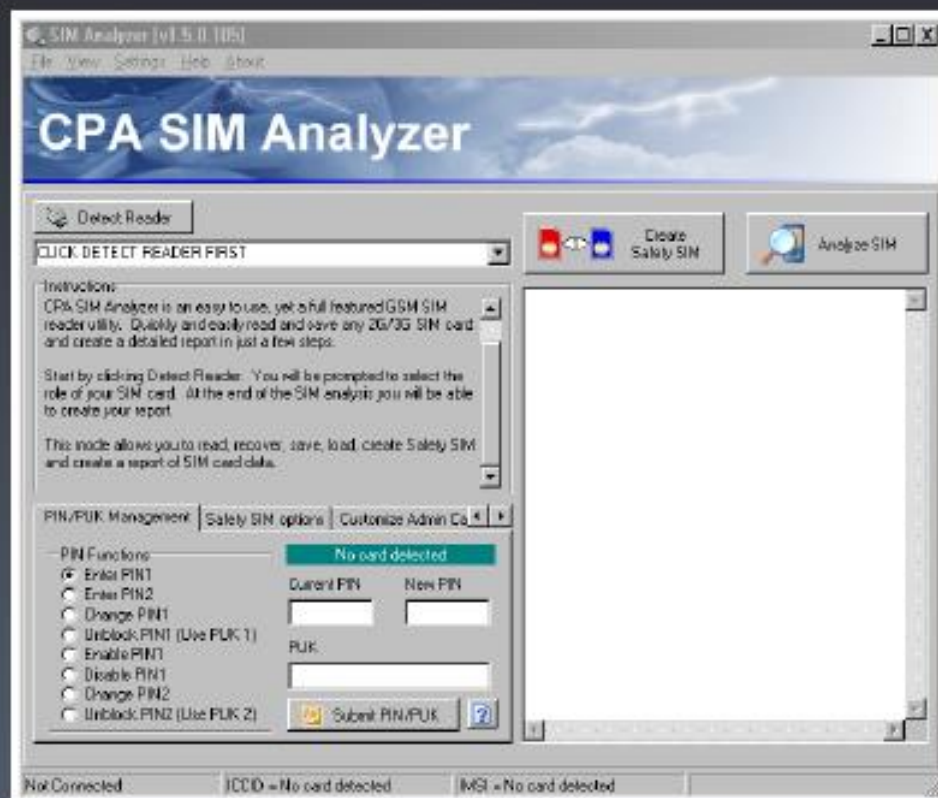
SIM Analyzer



SIM Analyzer is a cell phone forensics tool that **recovers the contents from SIM cards** of different mobile devices

It recovers:

- **Last Number Dialed, Abbreviated Dialing Numbers**
- **Active and Deleted text (SMS) messages**
- **All the general files found in the Telecom group as defined in the GSM 11.11v6 standards**

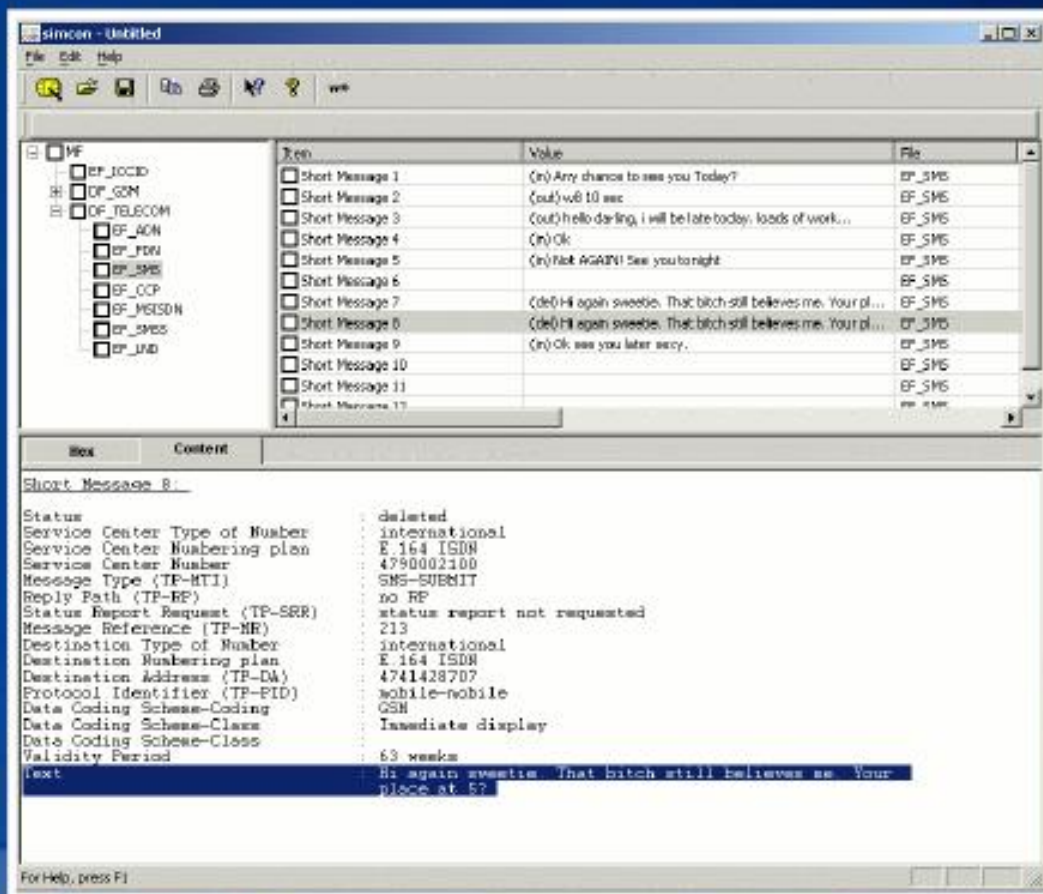


<http://www.bkforensics.com>

SIMCon

 SIMCon is a program that allows the user to **securely image all files** on a GSM/3G SIM card to a computer file with the SIMCon forensic SIM card reader

 It recovers deleted text **messages stored on the card** but not readable on phones



SIM Card Data Recovery

- ❗ SIM Card Data Recovery software **recovers accidentally deleted data** from mobile phone SIM cards

Data Doctor Recovery - SIM Card

Data Doctor Recovery SIM Card

Sim Card Data

- SMS (Text Messages)
- PhoneBook Number
- Last Dialed Number
- Fixed Dialed Number
- Service Provider Name
- Card Identification
- Location
- IMSI

S.No.	Sender Number	Date	Time	Message
1	+919910482928	14-11-07	15:57:55	[Inbox] This is a test
2	+919910482928	14-11-07	13:50:23	[Inbox] Hi!! Meeting you tomorrow!! Bye
3	55456	15-11-07	11:57:19	[Inbox UnRead] Pagein India ke cricket match ka score
4	+919910482928	14-11-07	14:25:28	[Inbox] Faithfulness is necessary for making relations...
5	+919910482928	14-11-07	14:28:10	[Inbox] Success is depend on hard work!! Remember as
6	+919910482928	14-11-07	14:29:20	[Inbox] Time and tide waits for none. So be on time wor
7	Find Friend	15-11-07	13:14:35	[Inbox UnRead] Ab aap bhi bana skate hain naye dost.
8	+919910482928	14-11-07	14:33:51	[Inbox] Wishing you happy birthday!! I need party...
9	+919910482928	14-11-07	14:36:39	[Inbox] I have got holidays for a week. I am coming to
10	+919910482928	14-11-07	14:39:08	[Inbox] Hi!! Talk to me immediately!! It is urgent!! Bye
11	Find Friend	15-11-07	14:20:53	[Inbox] Ab aap bhi bana skate hain naye dost. Try karke
12	+919910482928	14-11-07	14:49:32	[Inbox] I am celebrating my birthday party!! If you need to
13	+919910482928	14-11-07	14:50:22	[Inbox] Friend in need
14	+919910482928	14-11-07	14:53:23	[Inbox] I am studying for exams!! I will be talking to you...

Sender: +919910482928
Date: 14-11-07
Time: 14:28:10
Location: Inbox
Message: Success is depend on hard work!!
Remember and follow to succeed

<http://www.datadoctor.in>

Features

- ❗ Recovers **all SIM card details** missing due to accidental data deletion, human fault, and software or hardware malfunction
- ❗ Displays **SIM card IMSI number**, ICC Mobile Identification number, and Service provider name
- ❗ Supports all PC/SC standards or Phoenix standard SIM card reader used to **connect the SIM card** with the computer



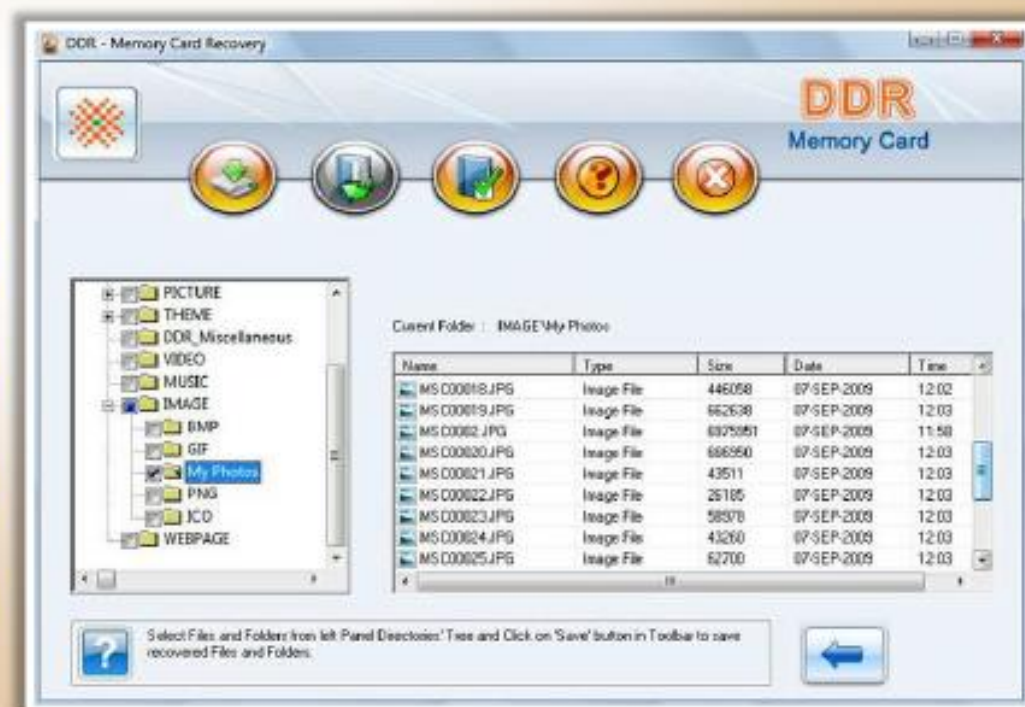
Memory Card Data Recovery

Memory Card Data Recovery **recovers** deleted pictures, lost images/photos, formatted audio/video files and folders, and encrypted data from the **corrupted memory card storage devices**



Features

- Retrieves deleted, erased or missing files and information from **corrupted or damaged memory cards**
- Supports easy **recovery of all media files** including image files, video files, audio files, text files, etc.
- Supports all **USB memory card** storage media devices including Compact Flash Memory card, Secure Digital SD, Multimedia Card, xD Picture card, Memory Stick, and other commonly used memory card storage media



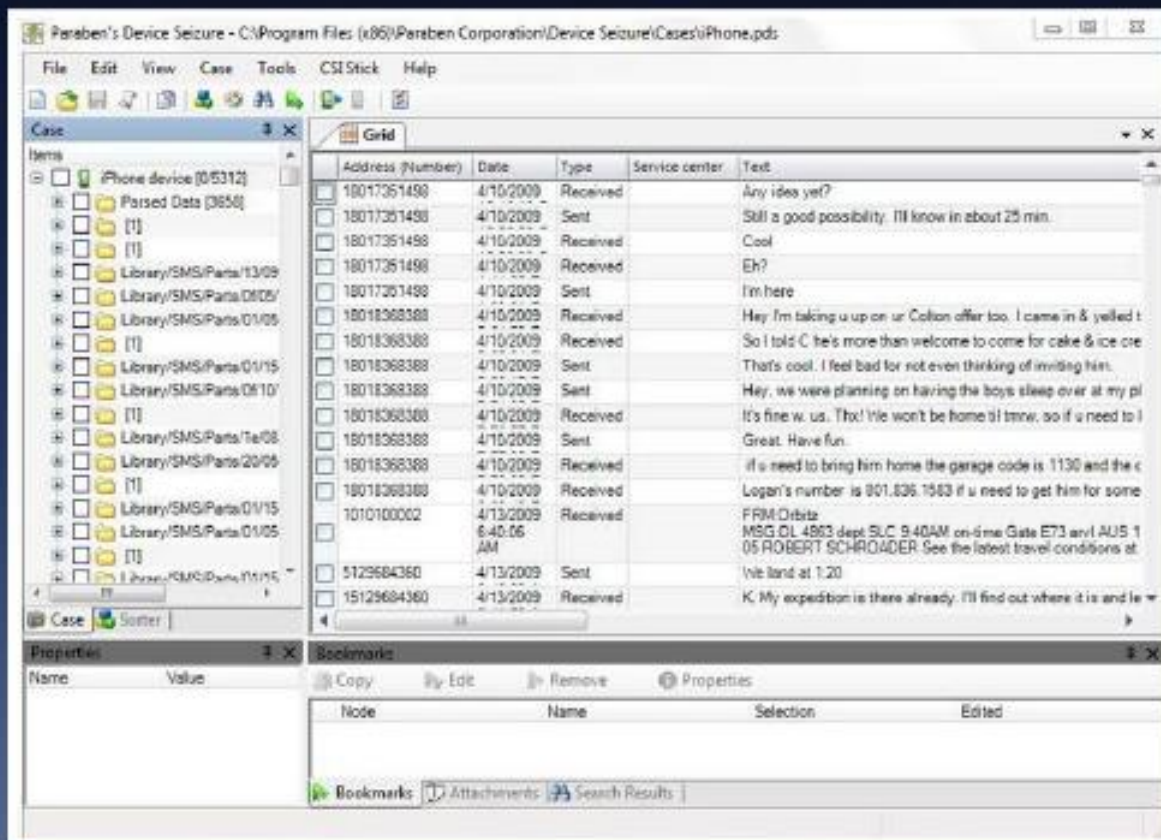
<http://www.datadoctor.in>

Device Seizure

Device Seizure is a **forensic acquisition and analysis tool** for examining cell phones, PDAs, and GPS devices

Features:

- Support of more than 1,900 devices
- Verification of file integrity using MD5 and SHA1 hash values
- Image viewing for graphic information, including data carving for multi-media files for most devices



<http://www.paraben.com>

SIM Card Seizure

SIM Card Seizure **recovers deleted SMS/text messages** and performs comprehensive analysis of SIM card data



Paraben's SIM Card Seizure

File Edit View Tools Help

Workspace

- SIM Card properties (1/43)
 - SIM IMEI (1/2)
 - IMEI (1)
 - Short messages (1/9)
 - Deliver SMS (deleted) (7)**
 - Binary data with 25 items (1)
 - SIM Abbreviated dialing numbers (1/2)
 - SIM Fixed dialing numbers (1/2)
 - SIM GSM specific (1/14)
 - SIM Lost number dialed (1/2)
 - SIM Service dialing Numbers (1/2)
 - SIM Telecom specific (1/9)

Grid

Record number	SCA	CA	Date/Time	
1	+19078319302	0000000000	2004-01-24 11:00:59 GMT-5	AW5.Care.atlwa.com00 YOUR A
2	+19078319302	1000000000	2004-01-25 00:20:30 GMT-5	3527352431
3	+19078319302	1000000000	2009-12-19 10:42:05 GMT-5	3529579947
4	+19078319302	1000000000	2009-12-19 12:50:13 GMT-5	3529579017
5	+19078319302	+13522172673	2009-12-20 14:35:55 GMT-5	WHERE IS THE WRAP
6	+19078319302	+13522172673	2009-12-20 14:42:12 GMT-5	I CAN ONLY TEXT MEACOUNT IS
7	+19078319302	1000000000	2009-12-04 12:31:59 GMT-5	3527352431

Properties

Name	Value
File	Deliver SMS (deleted)
Description	Deleted SMS Message
Deletion	WHERE IS THE WRAPPING...
Added	5/3/2005 3:31:13 PM
Edited	5/3/2005 3:31:13 PM

Bookmarks

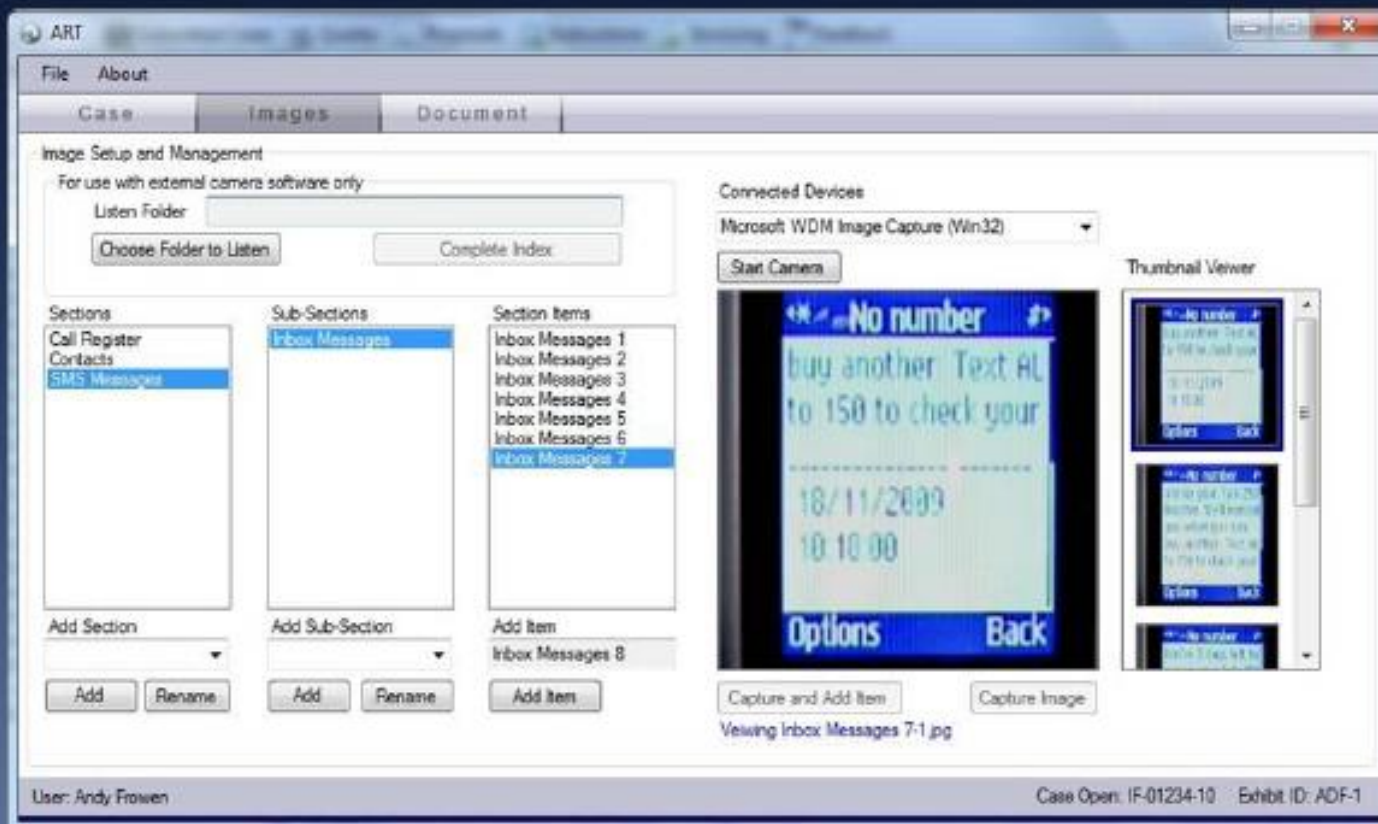
File	Name	Data	Timestamp
Deliver SMS (deleted)	Evidence 01	AW5.Care.atlwa.com#03	5/3/2005 3:30:41 PM
Deliver SMS (deleted)	Evidence 02	WHERE IS THE WRAPPING	5/3/2005 3:31:13 PM

Columns: 5, Row: 5

<http://www.paraben.com>

ART (Automatic Reporting Tool)

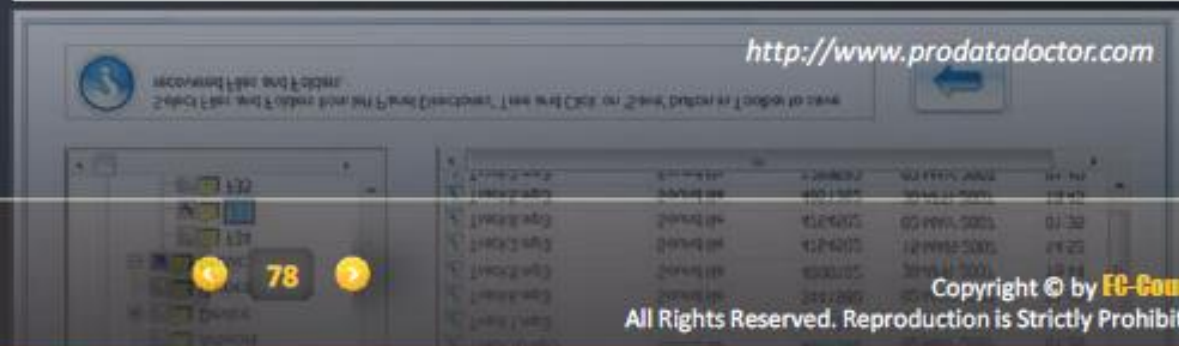
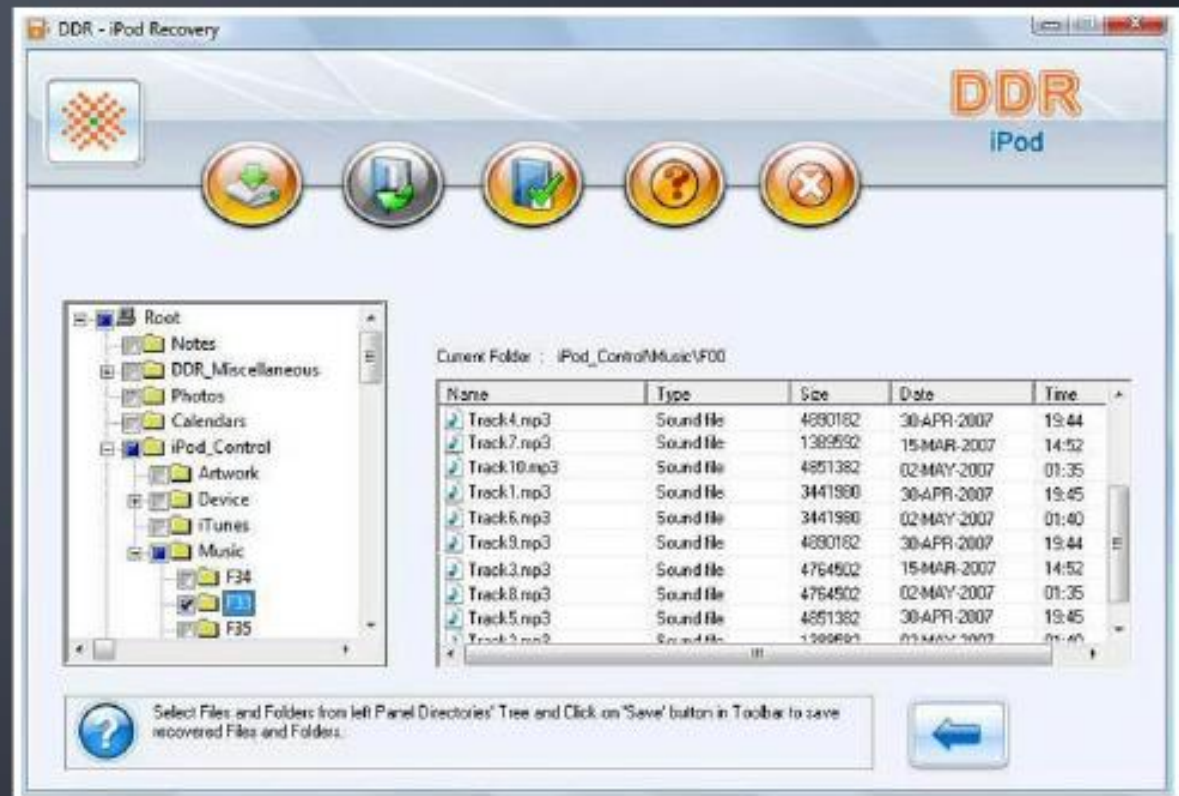
ART (Automatic Reporting Tool) is a software application developed to help Mobile Phone Forensic examiners to **capture images (via a USB camera) of mobile devices** and subsequently **produce a Microsoft Word document**



<http://www.intaforensics.com>

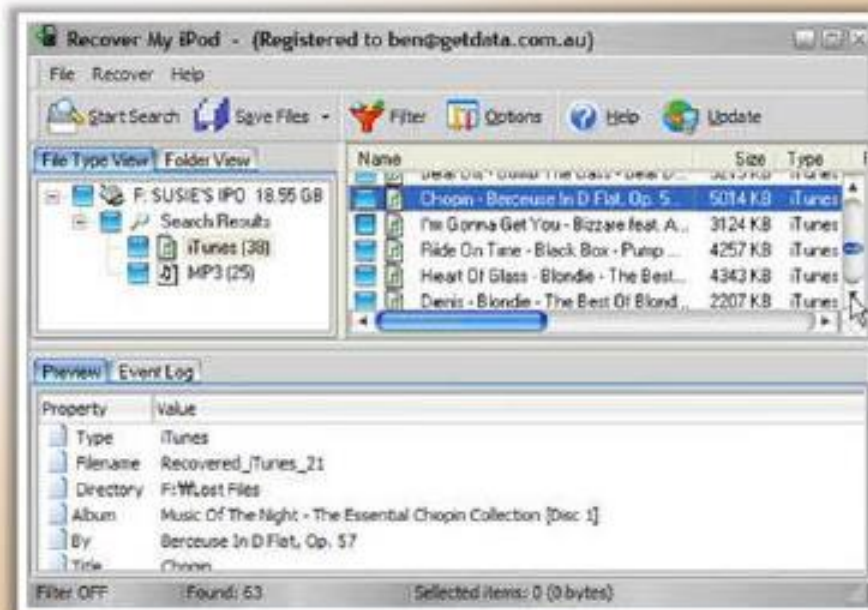
iPod Data Recovery Software

- iPod data recovery software easily **retrieves all lost deleted music files** from iPod digital media player
- It supports all **iPod models**, including iPod Classic, iPod Touch, iPod Mini, iPod Shuffle and other popular iPod audio-video models



Recover My iPod

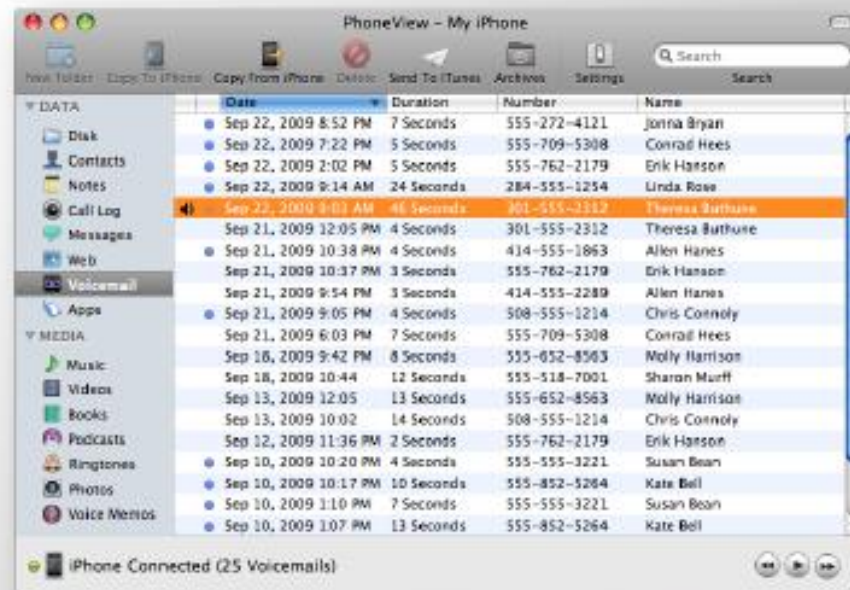
- Recover My iPod is iPod music recovery software to **recover deleted or lost iPod files**
- It recovers data after an **iPod Reset or Restore**
- It recovers data when an iPod gives a "Drive Not Formatted" message, or when an iPod is not recognized by a computer



<http://www.recovermyipod.com>

PhoneView

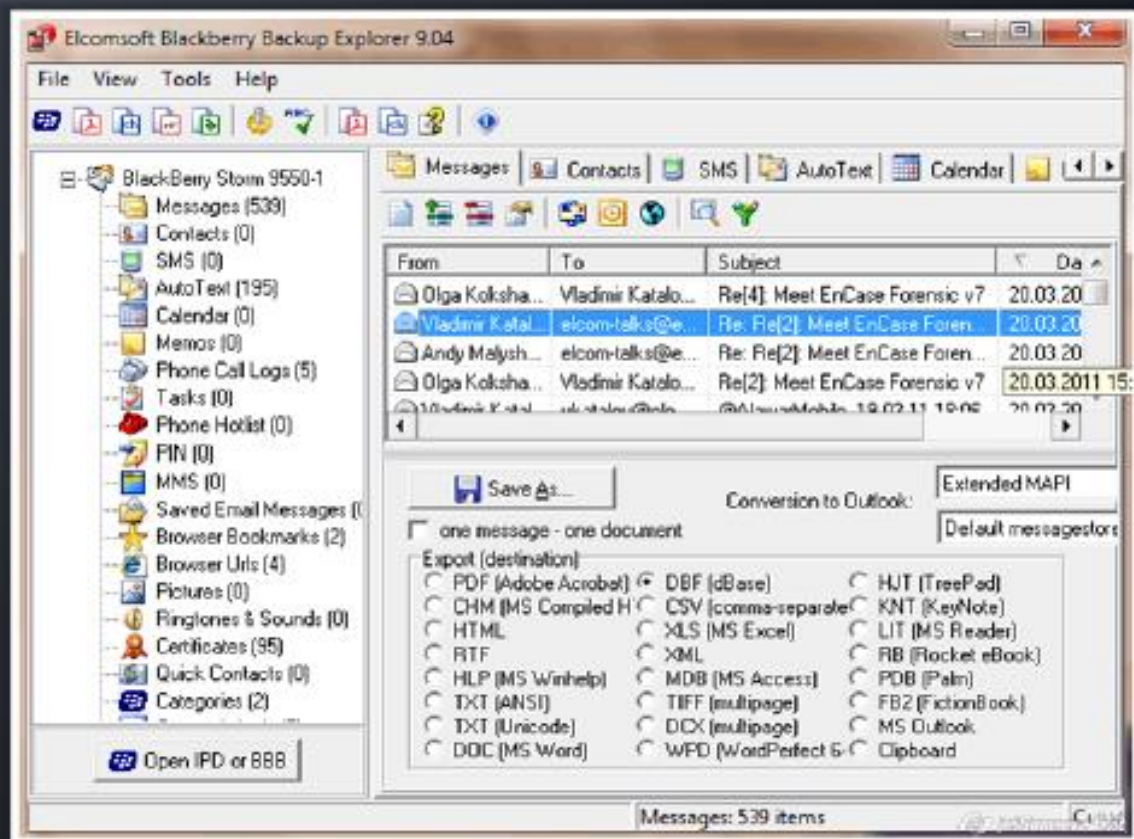
- PhoneView provides access to an **iPhone's media**, photos, notes, SMS messages, call history and contacts
- It automatically backs up messages from an iPhone and exports them as PDF, text, or XML files



<http://www.ecamm.com>

Elcomsoft BlackBerry Backup Explorer

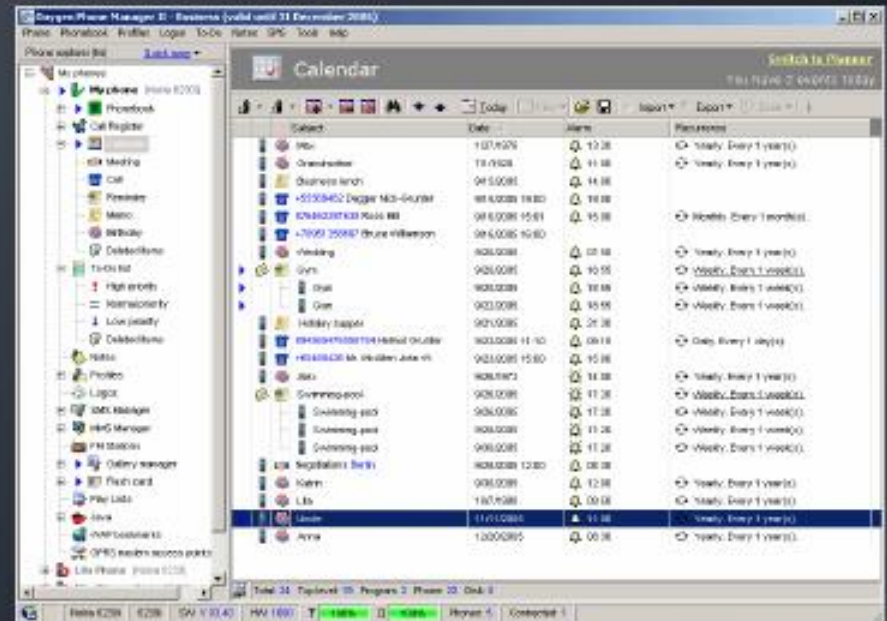
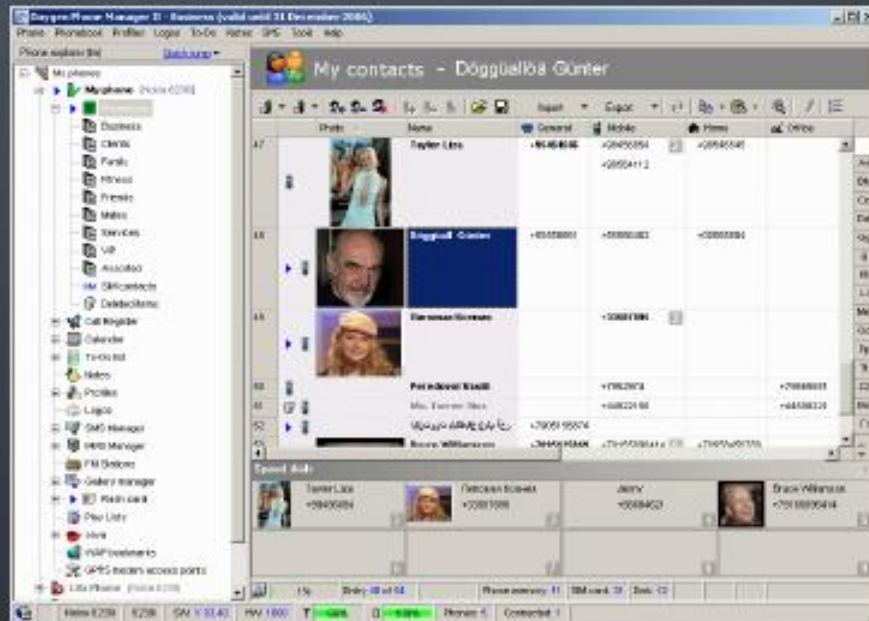
- Elcomsoft BlackBerry Backup Explorer assists **forensic specialists investigating the content of BlackBerry devices** by extracting, analyzing, printing or exporting the content of a BlackBerry backup produced with BlackBerry Desktop Software
- It supports a variety of **exporting options and formats** including PDF and HTML



<http://www.elcomsoft.com>

Oxygen Phone Manager II

- Oxygen Phone Manager II for Nokia phones provides a simple and convenient way to **control mobile phones from a PC**
- It **imports and exports data** from Windows Address Book and CSV files and saves data in various formats: MS Excel, MS Word, HTML, XML, etc.



<http://www.oxygensoftware.com>

Sanmaxi SIM Recoverer

- Sanmaxi SIM Recoverer recovers and restores deleted inbox, outbox or SMS messages from your mobile cell phone
- It can recover recently deleted text SMS messages and contacts



<http://www.sanmaxi.com>

Mobile Forensics Tools



USIMdetective

<http://quantaq.opiah.com>



Cell Phone Analyzer

<http://www.bkforensics.com>



CardRecovery

<http://www.cardrecovery.com>



iXAM

<http://www.ixam-forensics.com>



**Stellar Phoenix iPod
Recovery Software**

<http://www.stellarinfo.com>



**BlackBerry Database Viewer
Plus**

<http://www.cellica.com>



**iCare Data Recovery
Software**

<http://www.icare-recovery.com>



**BlackBerry Signing Authority
Tool**

<http://us.blackberry.com>

Module Flow



**Mobile
Phones**



**Mobile Operating
Systems**



**Mobile
Forensics**



**Mobile Forensics
Process**



**Mobile Forensics
Software Tools**



**Mobile Forensics
Hardware Tools**

Secure View Kit

- Secure View for Forensics is a software and hardware solution that **provides logical data extraction** of the content stored in a mobile phone
- It acquires cell phone data via **USB, Bluetooth, IrDA, and SIM card reader**

It acquires:

- Serial numbers:** IMEI (for GSM phones), and ESN (for CDMA) phones
- Recent calls:** Received calls, dialed calls & missed calls
- Contacts** (internal phone memory, as well as SIM card on supported GSM phones)
- Calendar** and to do lists
- Pictures and wallpapers
- Ring tones and music
- Video and movies



<http://www.datapilot.com>

Deployable Device Seizure (DDS)

- Deployable Device Seizure (DDS) is a version of Device Seizure designed for use in the field
- It is integrated into a tablet PC and is designed for one-click acquisitions of basic cell phone data such as call logs, address books, SMS messages, etc.



Paraben's Mobile Field Kit



<http://www.paraben.com>



Paraben's Mobile Field Kit is a **portable handheld forensic solution** that includes everything investigators need to perform a **comprehensive digital forensic analysis** of over 4,000 cell phones



Features:

- Performs **comprehensive analysis** to acquire data
- Examines various types of media such as **USB drives and SD Cards**
- **Recovers deleted data** with physical acquisitions
- Searches for **illicit images and chat logs** on computers



PhoneBase

- PhoneBase **extracts data** from any standard SIM card using a SIM Card Reader
- It **recovers the contents of SIM cards and phone memories**, including lists of phone numbers and associated names, recently made calls, and text messages



<http://www.phonebase.info>

XACT System



<http://www.forensicpeople.com>

- XACT enables you to perform “physical” **data investigations** of confiscated phones and allows **recovery of deleted information**
- It allows you to **acquire data** from locked phones and **deleted information**
- It **recovers deleted SMS** recovered from the SIM card and other information



Logicube CellDEK

- CellDEK is a **portable handset data extraction kit** designed for use at the scene of a crime and all working environments associated with ongoing investigations
- It can **access, read, and copy stored data** from GSM, CDMA, TDMA, iDen handsets, SIM cards, PDAs, and 15 types of flash cards



Features

- Extracts handset time and date**, serial numbers (IMEI, IMSI), dialed calls, missed calls, received calls, phonebook (both handset and SIM), SMS (both handset and SIM), deleted SMS from SIM, calendar, memos, and to do lists
- Built-in SIM card reader** and SIM card-reading software
- Connection and control** of external jammer to prevent loss of data
- Time-stamped forensic audit trail** records data sent and received from target device



<http://www.logicubeforensics.com>

Logicube **CellDEK TEK**

The CellDEK® TEK is designed for the **technically experienced forensic investigator** who requires a comprehensive, compact and portable data extraction hardware solution for mobile devices

It is **compatible with 2000 of the most popular cell phones** and PDAs, including Blackberry devices

It **extracts data** from Apple iPhone, iPhone 3G, and iPod touch devices as well as Garmin, TomTom, and iPAQ **satellite navigation devices**



<http://www.logicubeforensics.com>

RadioTactics **ACESO**

- RadioTactics ACESO recovers **digital evidence** from mobile phones, SIM cards and media cards
- It allows investigators to **examine and interrogate phones** and other mobile devices, **quickly and accurately**



<http://www.radio-tactics.com>

Features:

- **Provides critical details**, including time and date, geotagging and make and model of device used
- **Safely block network access** using the patent pending Handset Access Card
- **Capture data from devices** such as sat navs and flash media



UME-36Pro - Universal Memory Exchanger



Cellebrite's UME-36Pro is a **phone memory transfer and backup solution**

Supports transfer of **content** across all mobile handset technologies



Transfer of phones' internal memory and SIM card content with integrated SIM/Smart Card reader



Transfer, backup, and restoration of mobile phone content



<http://www.cellebrite.com>

Cellebrite UFED System - Universal Forensic Extraction Device

Using OpenSSH you can **tunnel all of the traffic from your local box to a remote box** that you have an account on



<http://www.cellebrite.com>



It **extracts data** from all cell phones: phonebooks, pictures, videos, text messages, call logs, ESN and IMEI information



It is a **standalone kit**, with no computer required for extraction



It generates complete, **MD5 verified evidence reports**



It supports over **1,400 handset models**

ZRT 2

- ZRT 2 is a **cell phone forensic investigation solution that supports all phones** and can be used on its own or in conjunction with existing tools

Features:

- It completely streamlines the process of taking **high-resolution photographs** of screen displays
- It **merges photos** into custom designed report templates



<http://www.fernico.com>

ICD 5200



- Paraben's Project-A-Phone™ ICD-5200 is a cell phone screen capture device that **allows you to take pictures or videos** of the screen of almost any cell phone and display them right on your computer
- It **captures the display screen** at up to 5.2 megapixel resolution

Features:

- It captures **evidence** in cell phone forensics
- It assists at **live meetings** where you want to present from a computer
- It provides **web-based** demonstrations
- It can take **screen shots** for print marketing materials or documentation



<http://www.projectaphone.com>

ICD 1300



<http://www.projectaphone.com>

- ICD 1300 is a Project-a-Phone product designed for **forensic investigations of cell phones**
- It captures the **display screen** at up to 1.3 megapixel resolution

Features:

- It records forensic evidence
- It offers **screenshots** for digital marketing materials or documentation



Module Summary



- ☐ A mobile phone or cellular phone is an electronic device used for mobile voice or data communication over a network
- ☐ A mobile operating system is an operating system that operates a mobile device like a mobile phone, smart phone, PDA, etc.
- ☐ Mobile phone forensics is the science of recovering digital evidence from a mobile phone under forensically sound conditions
- ☐ Collect and preserve all the electronic devices found at the crime scene
- ☐ The data acquisition process at the crime scene is hampered due to the lack of controlled settings, appropriate equipment, and other prerequisites
- ☐ Mobile Forensic software collects all possible data from a mobile phone and generates an extensive forensics report